Estero Bay Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators **2**

Nutrients . . . 2

Total Nitrogen - Discrete . . . 2

Total Phosphorus - Discrete . . . 4

Water Quality . . . 6

Dissolved Oxygen - Continuous . . . 6

Dissolved Oxygen - Discrete . . . 8

Dissolved Oxygen Saturation - Continuous . . . 10

Dissolved Oxygen Saturation - Discrete . . . 12

Salinity - Discrete . . . 14

Salinity - Continuous . . . 16

Water Temperature - Continuous . . . 18

Water Temperature - Discrete . . . 20

pH - Continuous . . . 22

pH - Discrete . . . 24

Specific Conductivity - Continuous . . . 26

Water Clarity . . . 28

Turbidity - Continuous . . . 28

Turbidity - Discrete . . . 30

Total Suspended Solids - Discrete . . . 32

Chlorophyll a, Uncorrected for Pheophytin - Discrete . . . 34

Chlorophyll a, Corrected for Pheophytin - Discrete . . . 36

Secchi Depth - Discrete . . . 38

Colored Dissolved Organic Matter - Discrete . . . 40

Indicators

Nutrients

Total Nitrogen - Discrete

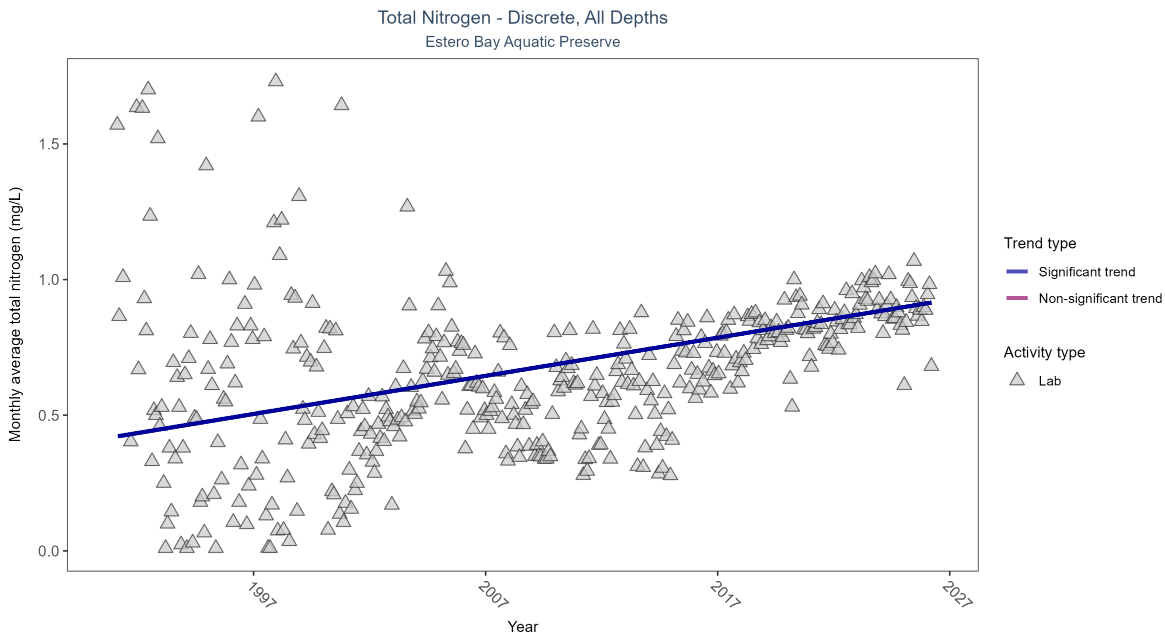

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	8330	36	1991 - 2026	0.658	0.35259	0.42027	0.01406	0

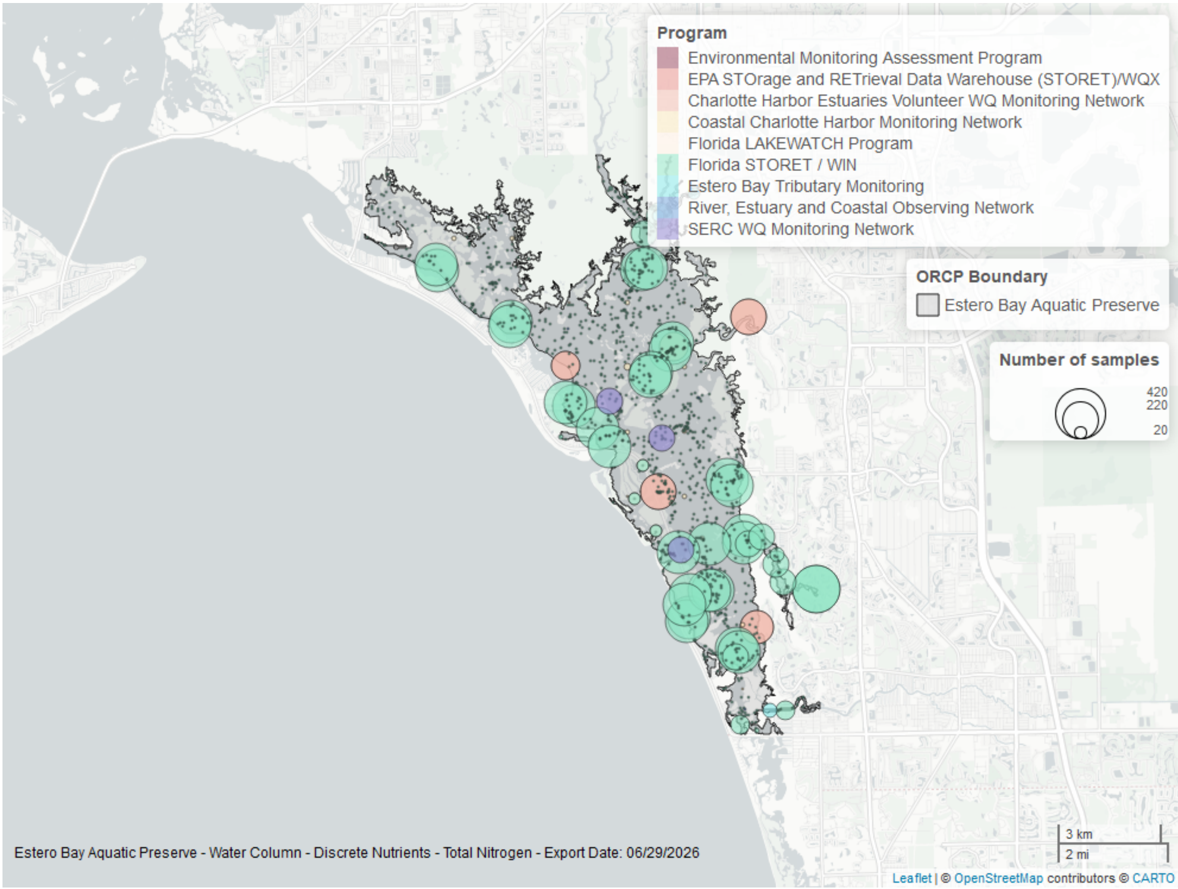

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

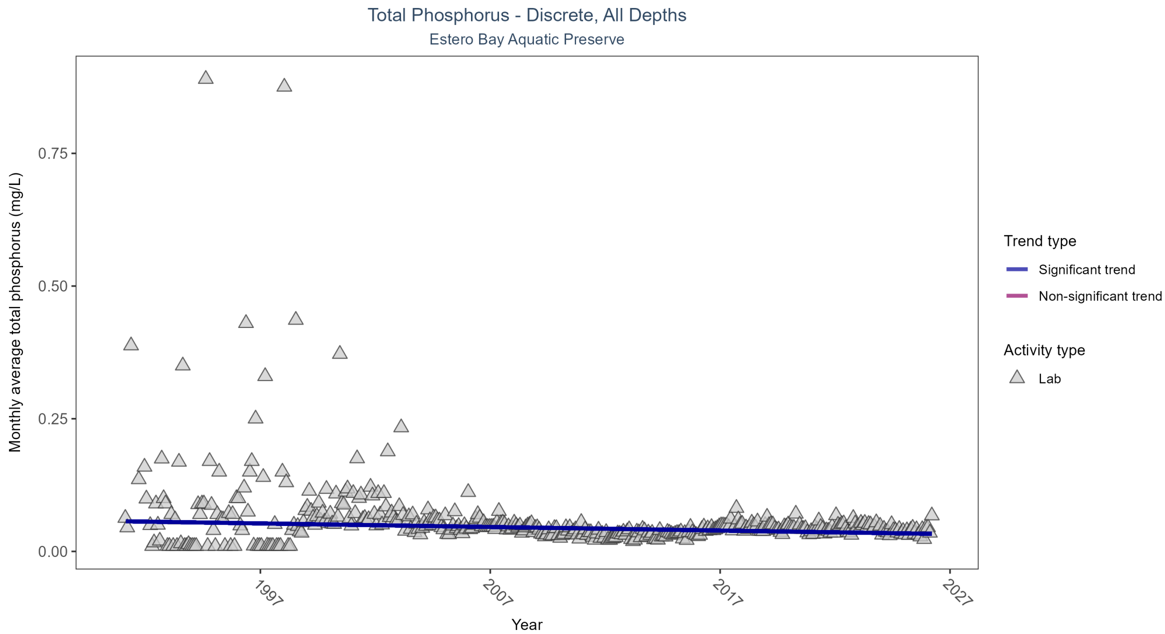

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	8335	36	1991 - 2026	0.041	-0.19849	0.05679	-0.00067	0

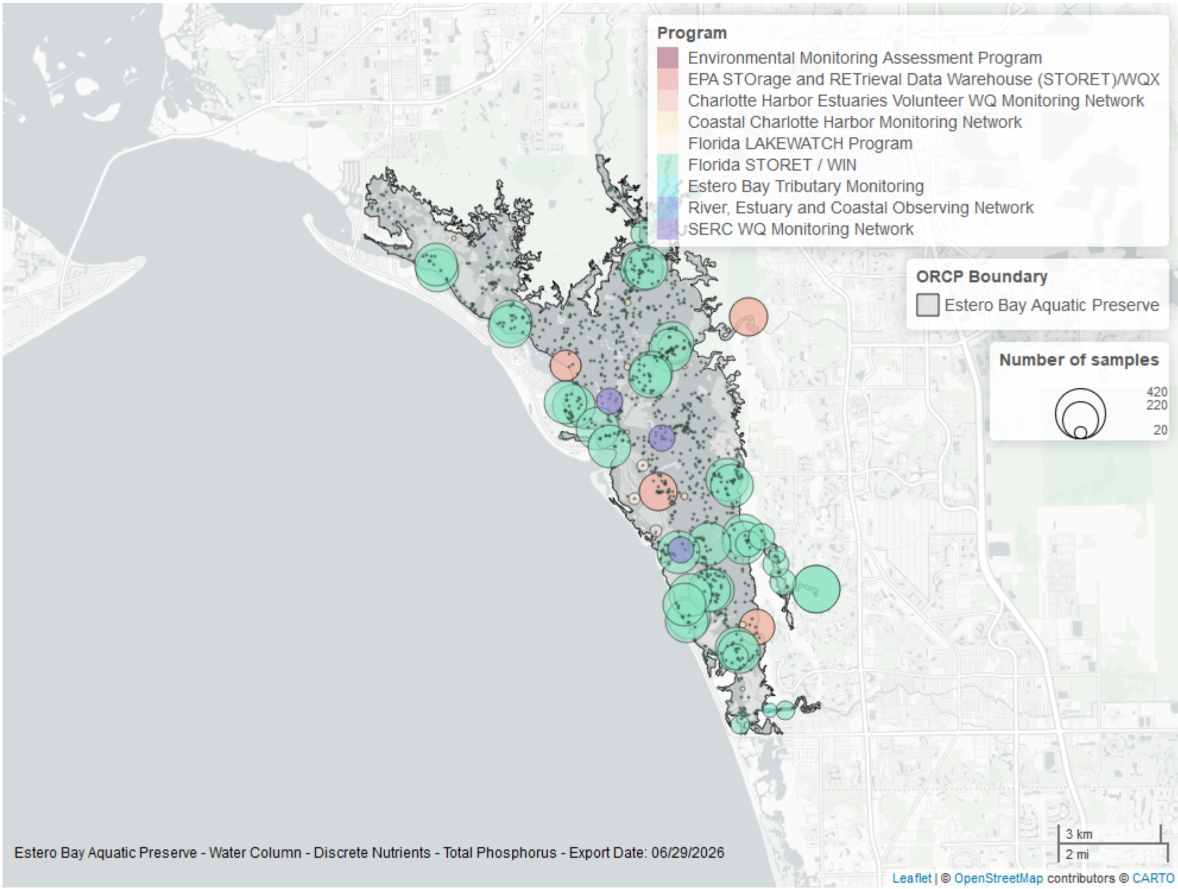

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

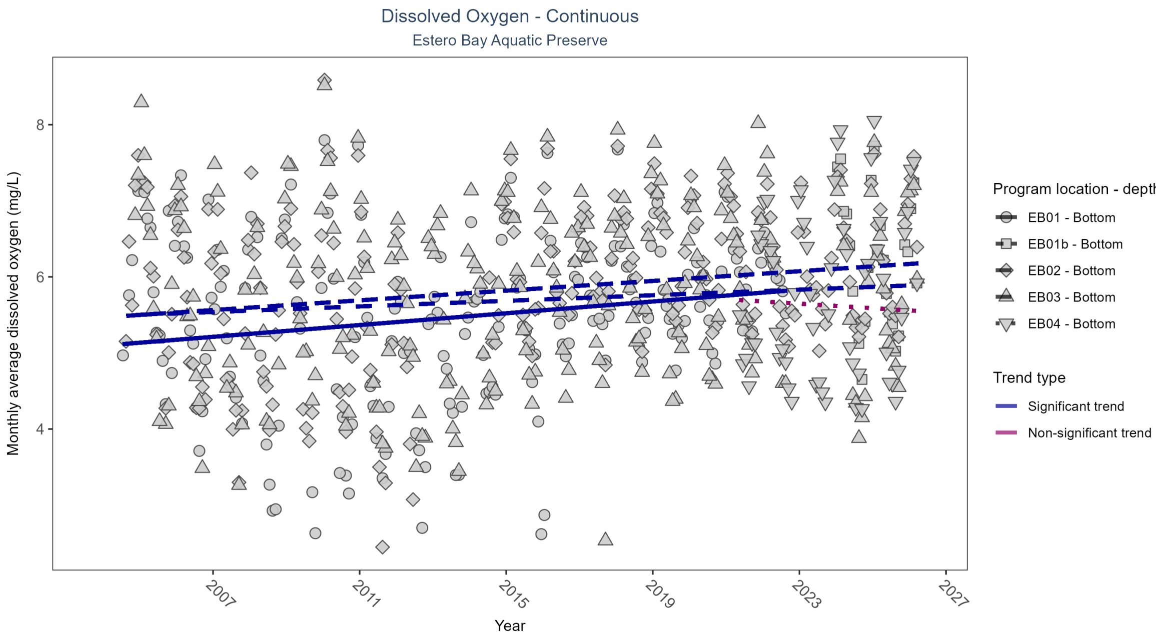

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At three program locations, monthly average dissolved oxygen increased between 0.02 and 0.04 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Increasing trend	478415	19	2004 - 2022	5.6	0.22	5.1	0.04	0
EB01b	Insufficient data	74584	3	2024 - 2026	6.2	-	-	-	-
EB02	Increasing trend	521982	22	2004 - 2026	6.1	0.25	5.47	0.03	0
EB03	Increasing trend	514210	22	2004 - 2026	5.9	0.15	5.48	0.02	0
EB04	No detectable trend	159166	6	2021 - 2026	5.8	-0.03	5.71	-0.03	0.83

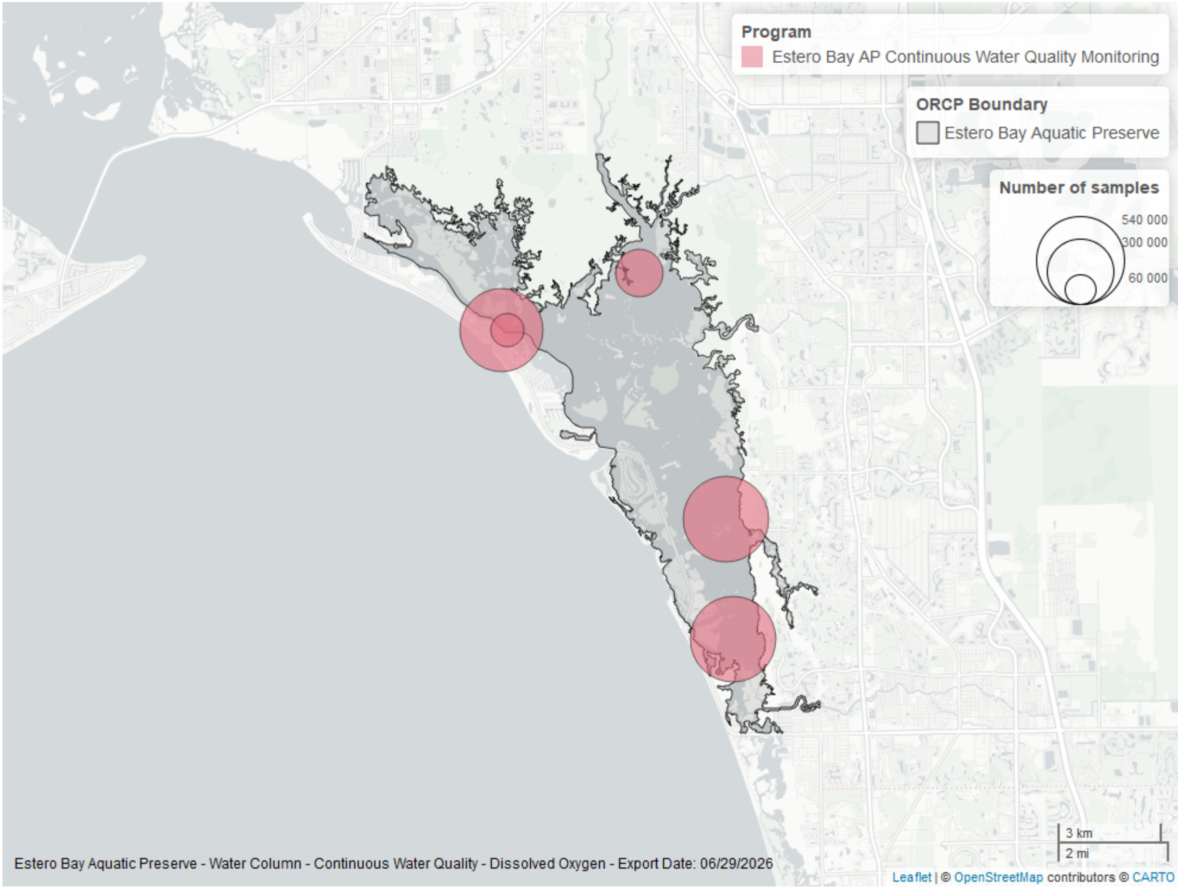

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

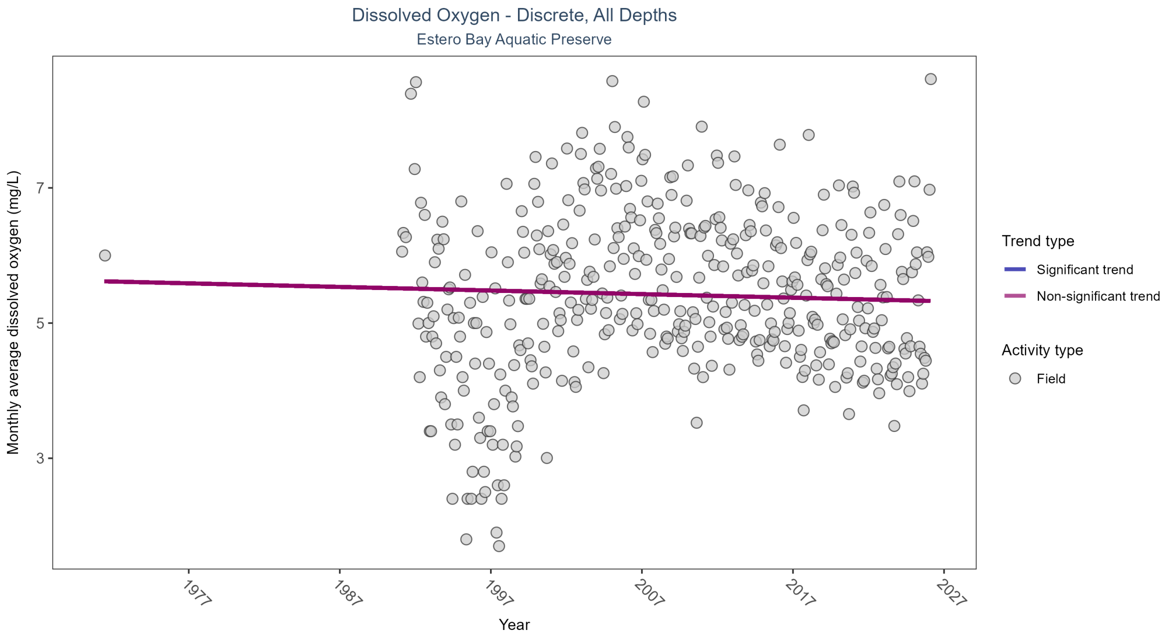

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1971 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	11787	37	1971 - 2026	5.8	-0.04521	5.61967	-0.00531	0.2062

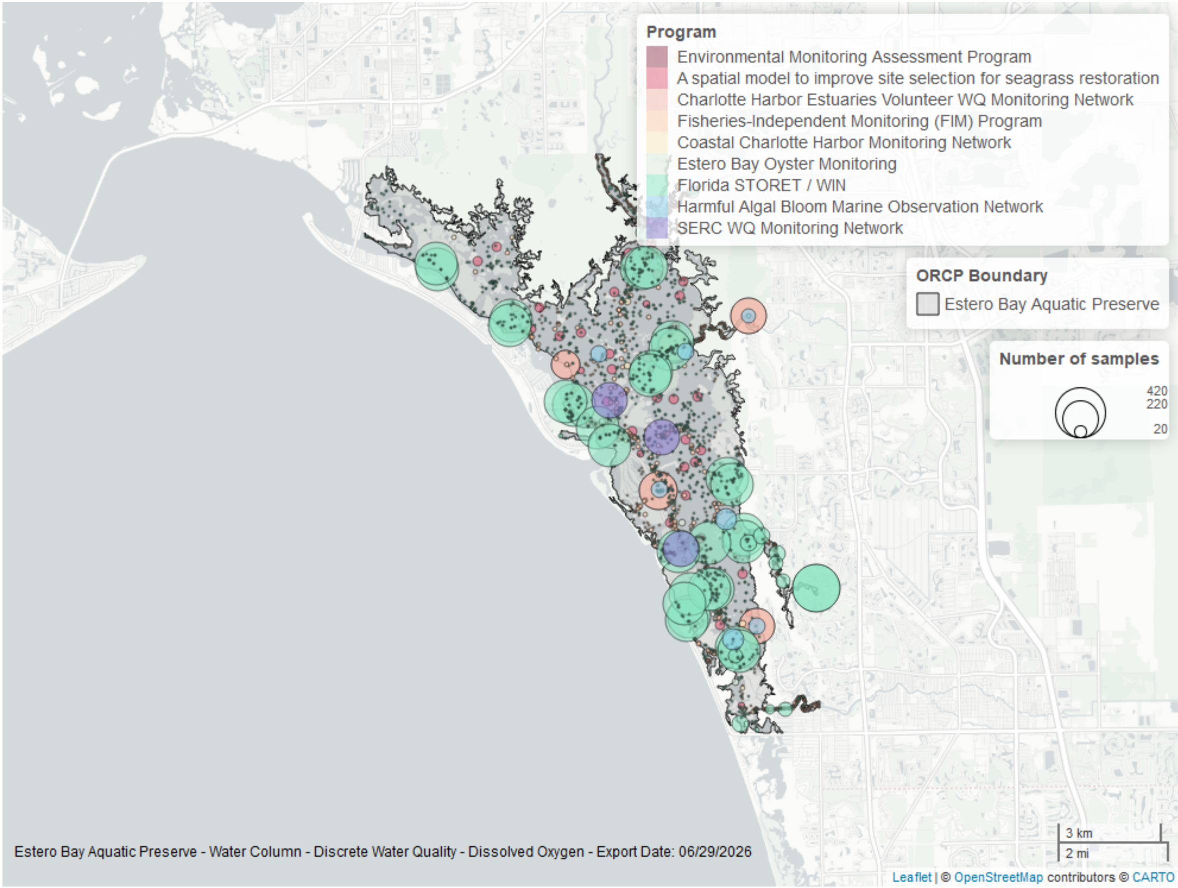

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

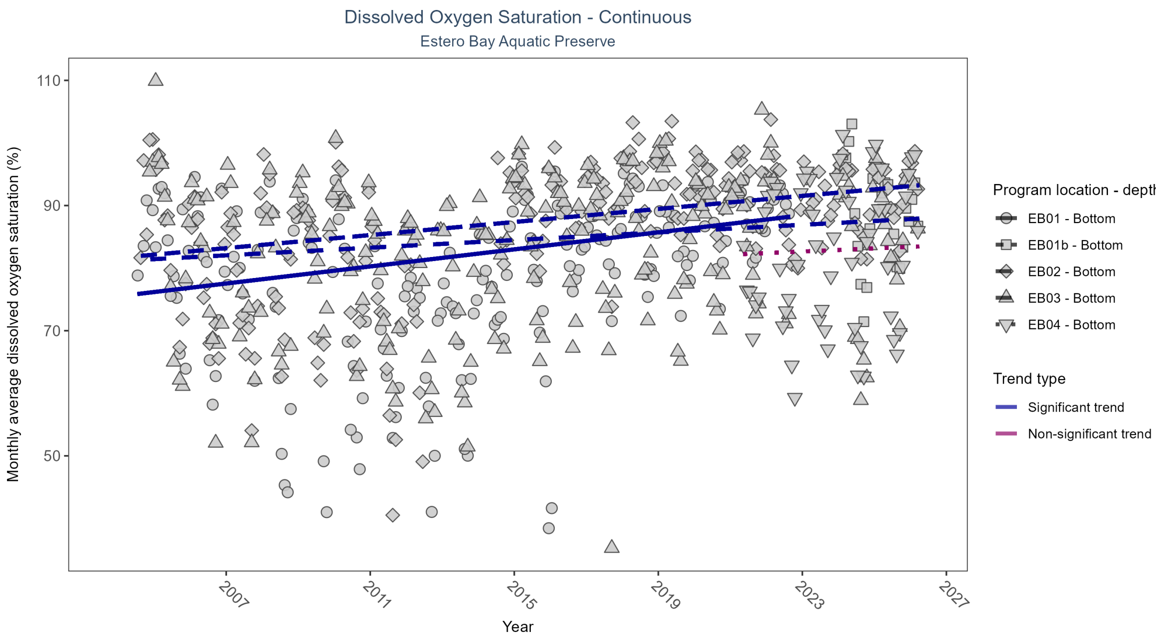

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At three program locations, monthly average dissolved oxygen saturation increased between 0.3 and 0.68% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Increasing trend	480233	19	2004 - 2022	81.6	0.31	75.5	0.68	0
EB01b	Insufficient data	75294	3	2024 - 2026	88.7	-	-	-	-
EB02	Increasing trend	524149	22	2004 - 2026	88.4	0.36	81.63	0.52	0
EB03	Increasing trend	517535	22	2004 - 2026	84.0	0.21	81.13	0.3	0
EB04	No detectable trend	163658	6	2021 - 2026	84.1	0.06	82.16	0.25	0.61

\end{table}

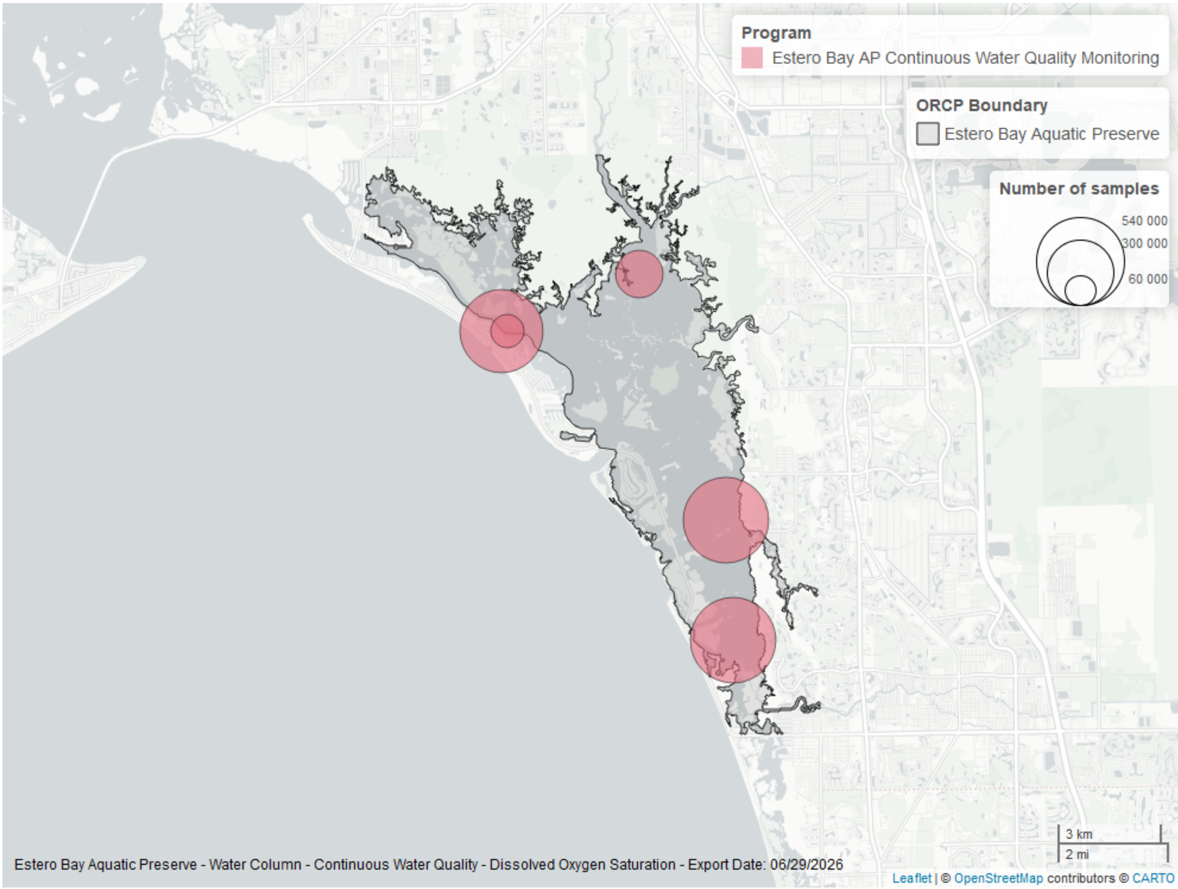

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

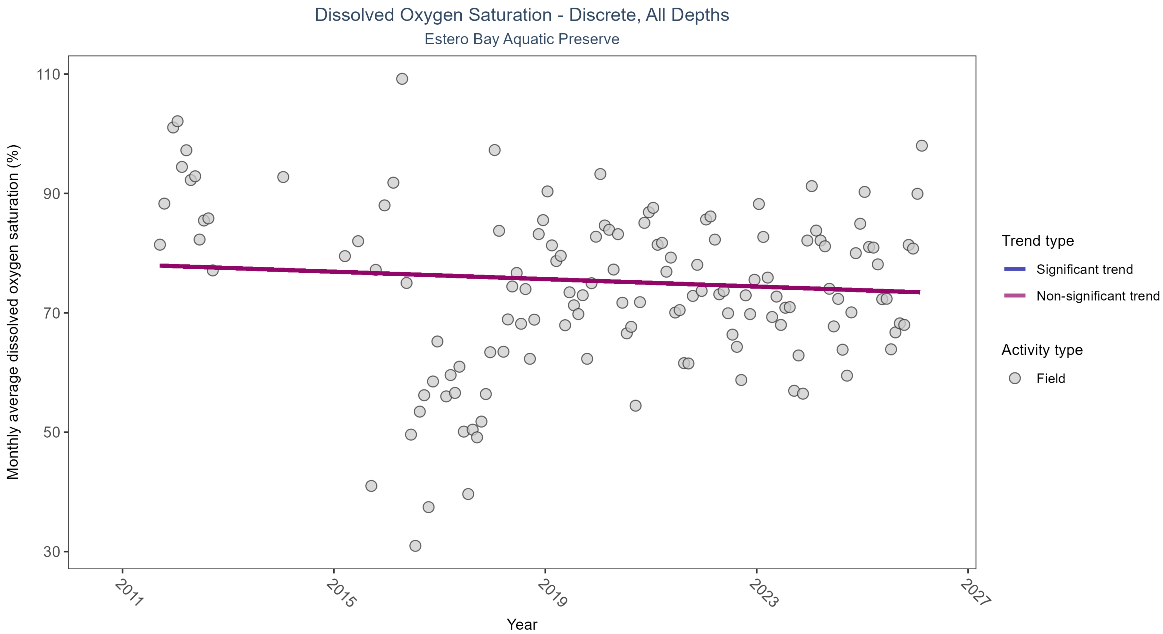

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 5: Dissolved oxygen saturation showed no detectable trend between 2011 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3141	15	2011 - 2026	81.9	-0.09958	78.12872	-0.31054	0.1277

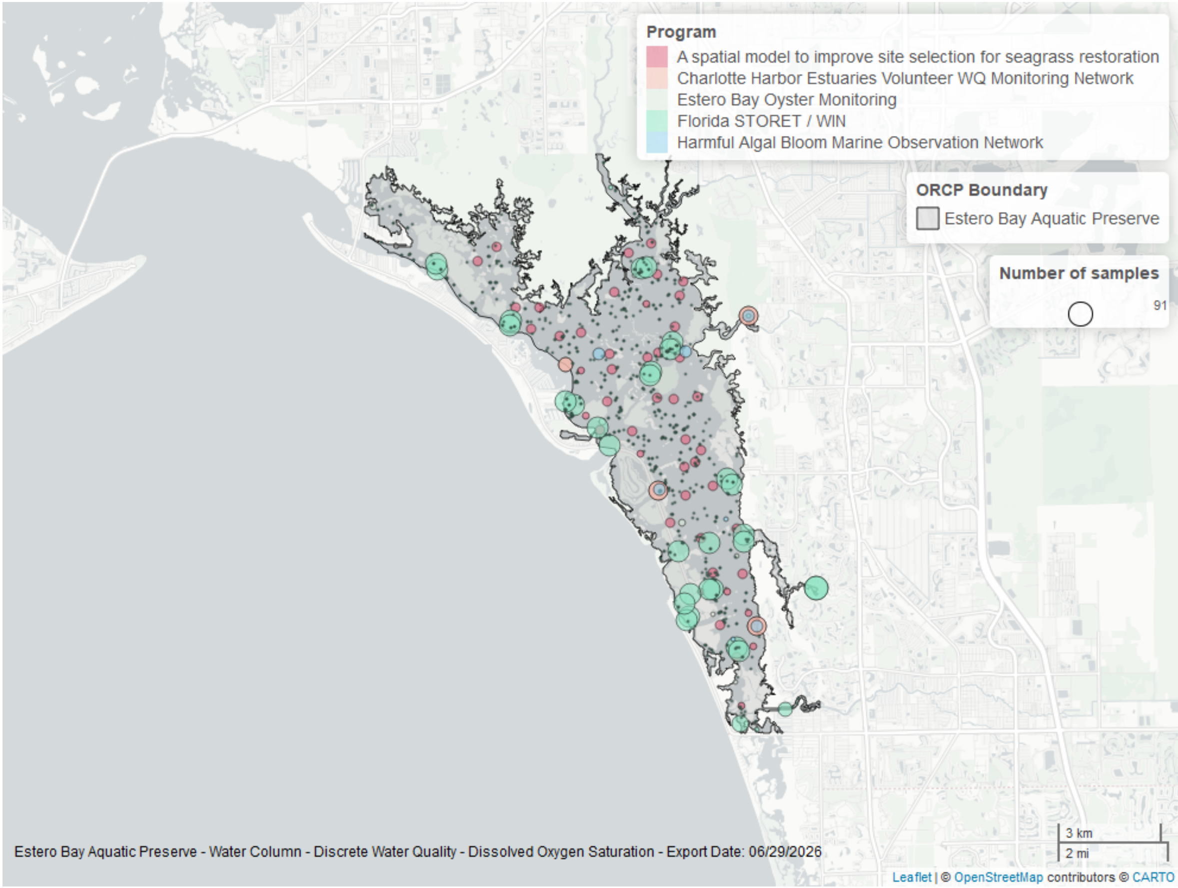

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

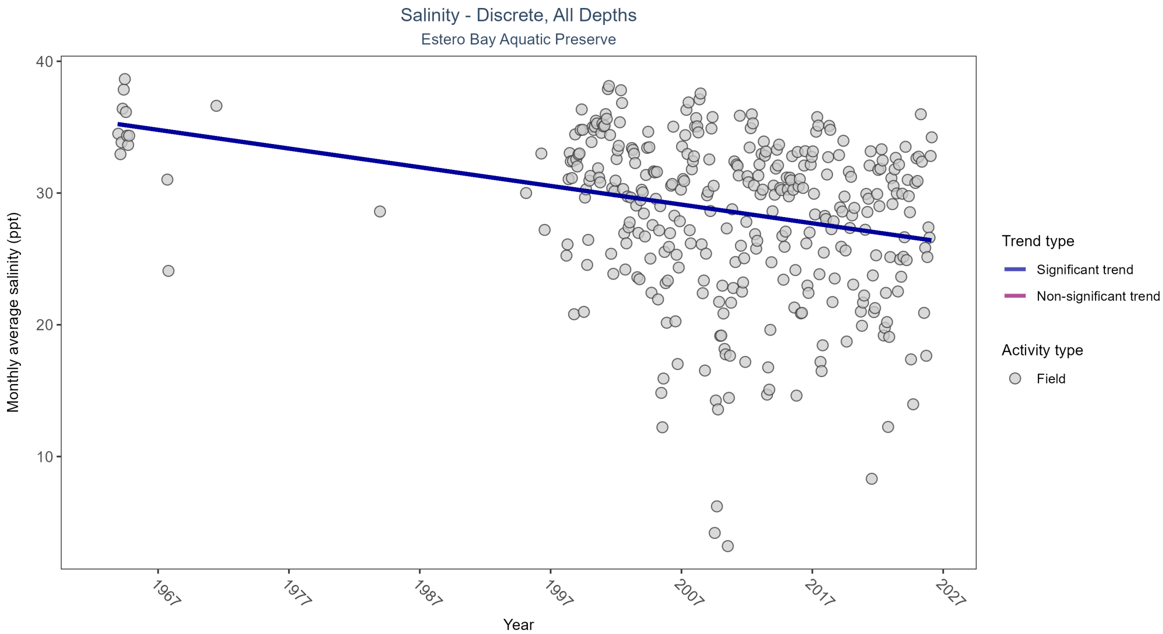

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Monthly average salinity decreased by 0.14 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	5397	36	1963 - 2026	32.2	-0.25714	35.36244	-0.14182	0

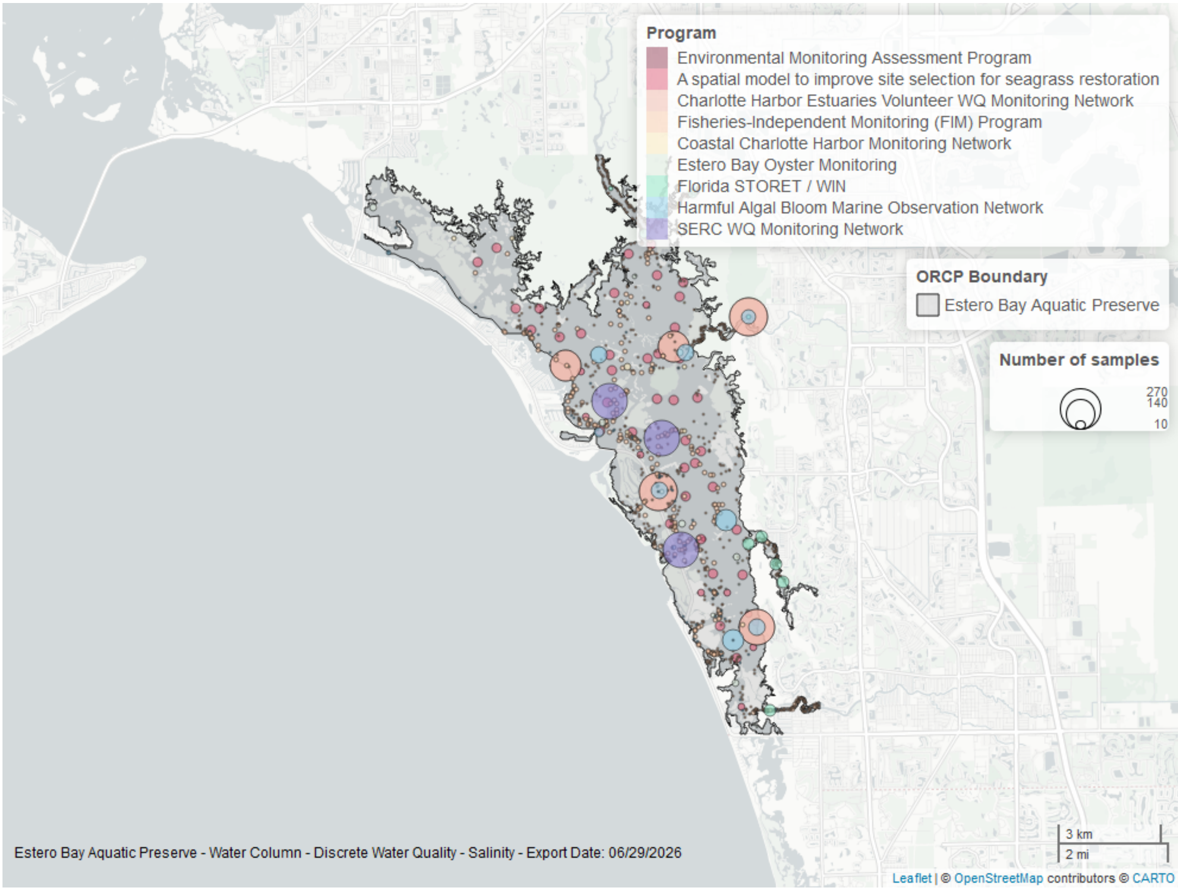

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

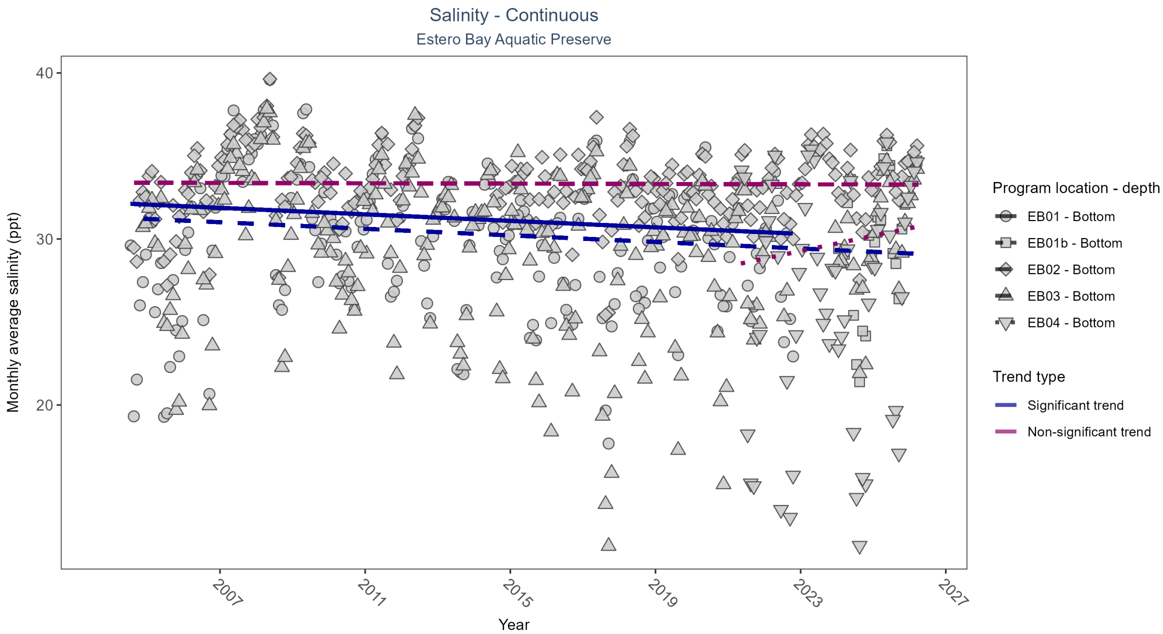

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At two program locations, monthly average salinity decreased by 0.1 ppt per year. No detectable change in monthly average salinity was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Decreasing trend	567163	19	2004 - 2022	30.7	-0.13	32.17	-0.1	0.01
EB01b	Insufficient data	70428	3	2024 - 2026	30.5	-	-	-	-
EB02	No detectable trend	612796	22	2004 - 2026	33.7	-0.01	33.39	-0.01	0.81
EB03	Decreasing trend	606608	23	2004 - 2026	30.9	-0.15	31.31	-0.1	0
EB04	No detectable trend	160786	6	2021 - 2026	28.2	0.15	28.36	0.46	0.21

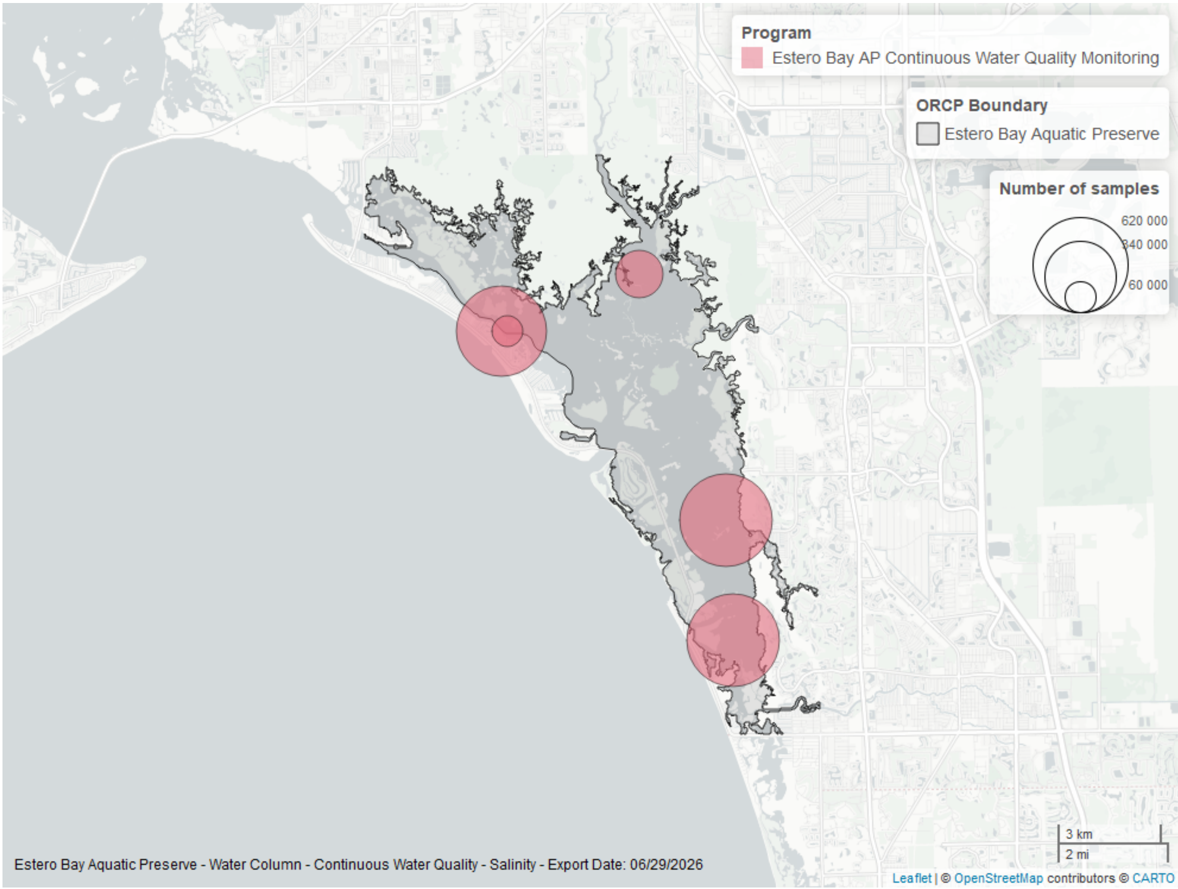

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

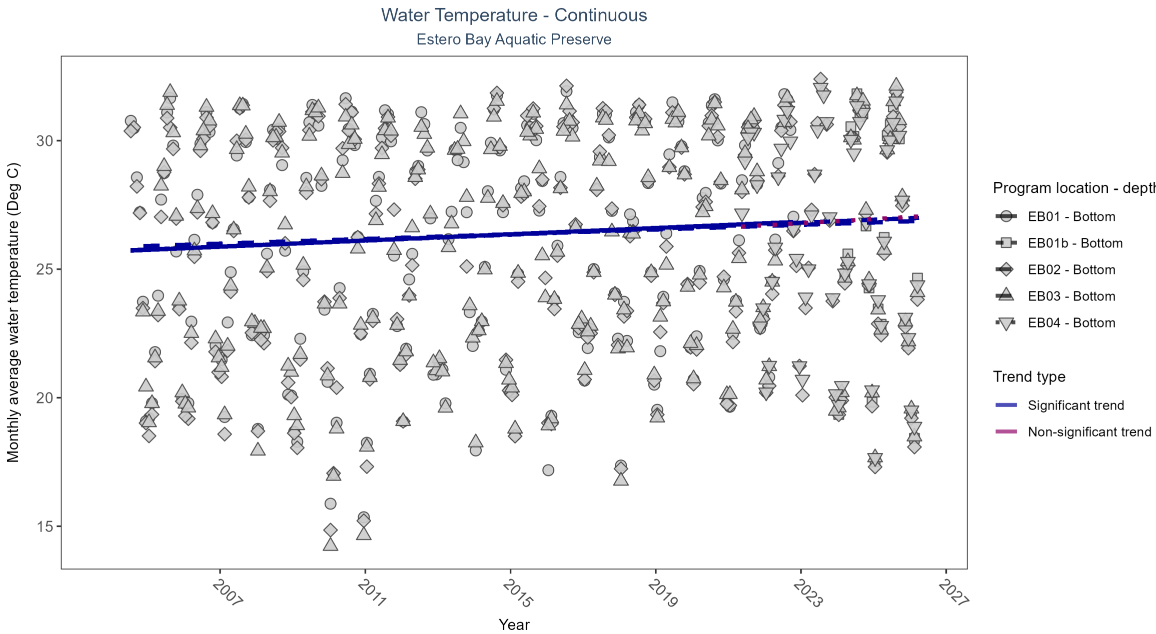

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At three program locations, monthly average water temperature increased between 0.05 and 0.06Deg C per year. No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Increasing trend	617636	19	2004 - 2022	26.8	0.25	25.69	0.06	0
EB01b	Insufficient data	75287	3	2024 - 2026	26.2	-	-	-	-
EB02	Increasing trend	661155	23	2004 - 2026	26.4	0.29	25.71	0.06	0
EB03	Increasing trend	651183	23	2004 - 2026	26.4	0.19	25.86	0.05	0
EB04	No detectable trend	170819	6	2021 - 2026	26.9	0.03	26.63	0.08	0.83

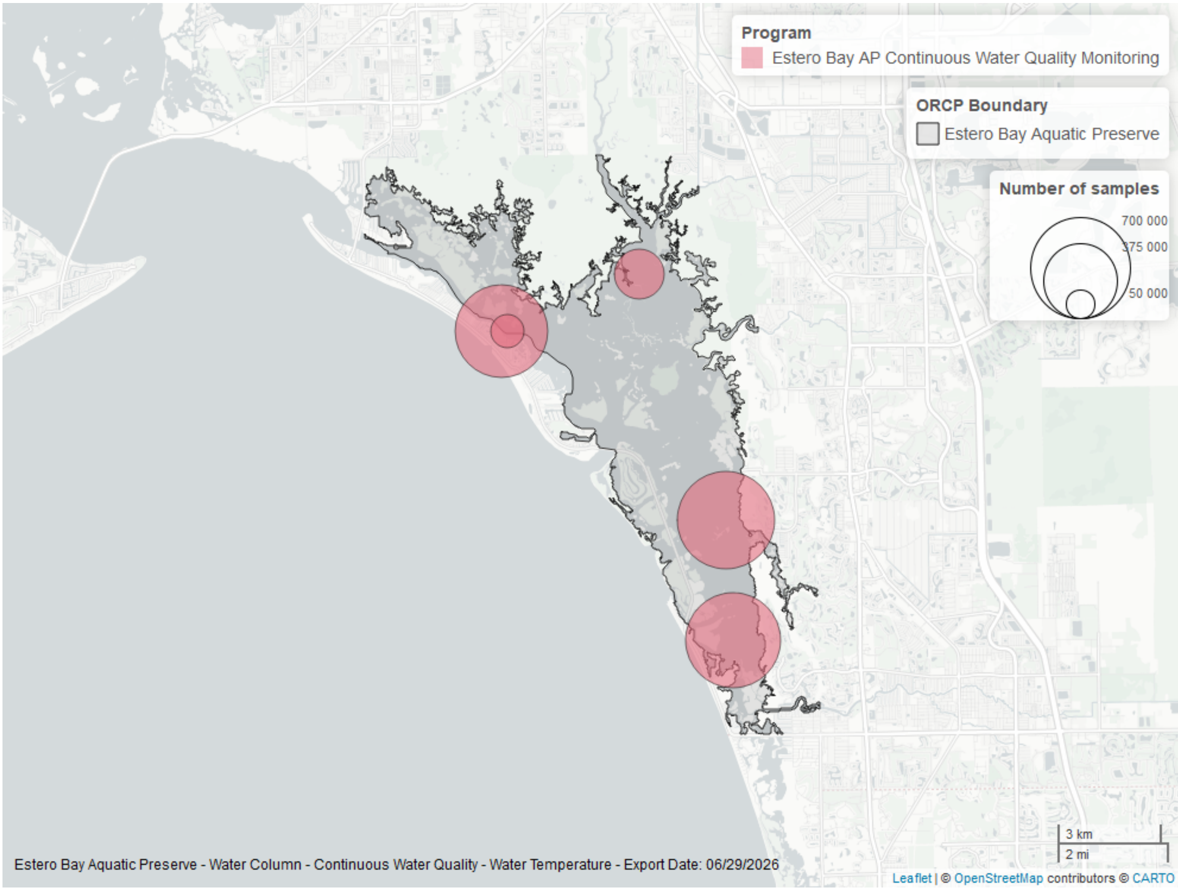

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

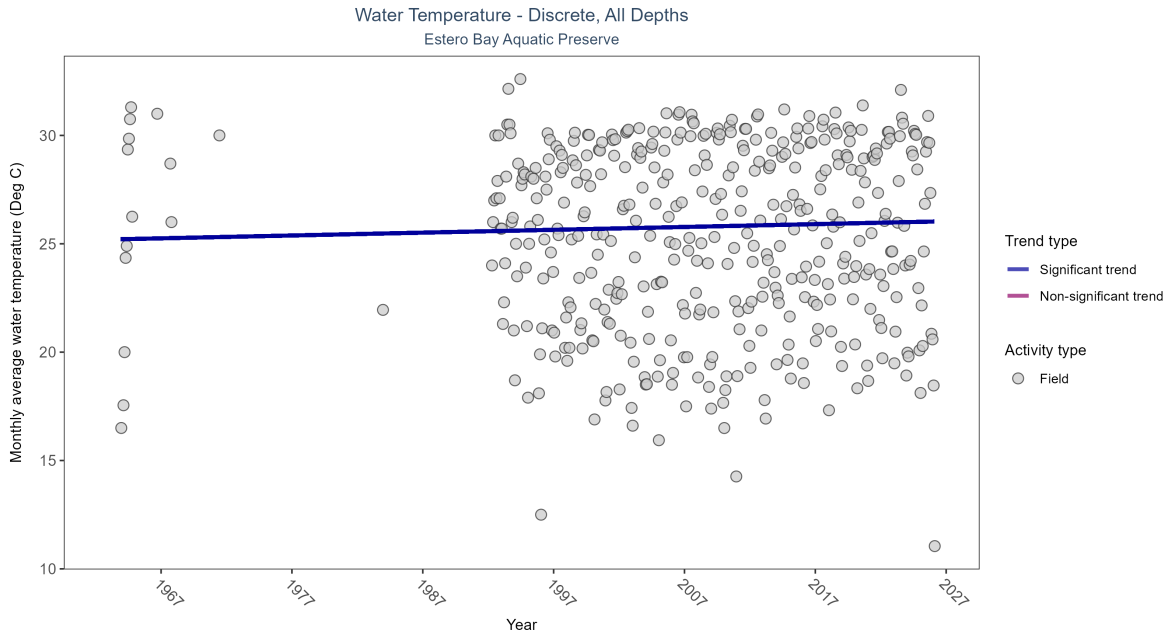

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Monthly average water temperature increased by 0.01Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	11428	41	1963 - 2026	25.9	0.07638	25.19902	0.01311	0.0245

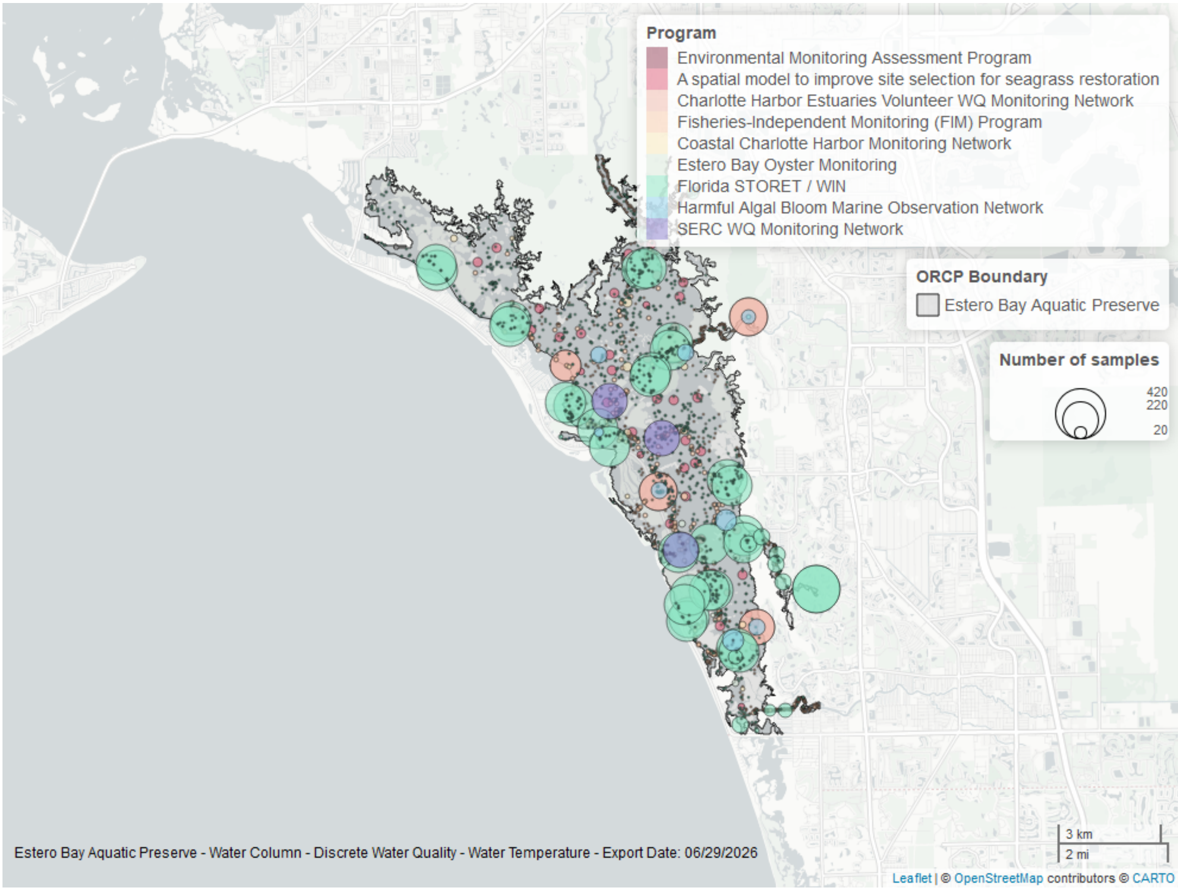

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

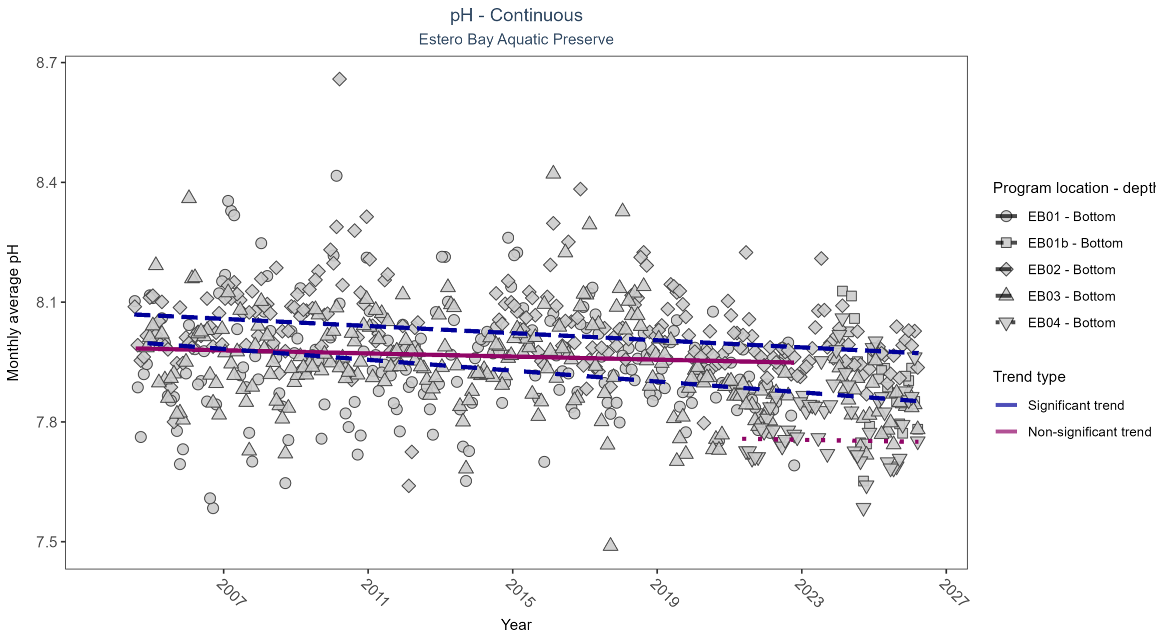

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At two program locations, monthly average pH decreased by less than 0.01 pH units per year at one site and by 0.01 pH units per year at the other. No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	No detectable trend	562188	19	2004 - 2022	7.9	-0.08	7.99	0	0.12
EB01b	Insufficient data	71881	3	2024 - 2026	7.9	-	-	-	-
EB02	Decreasing trend	596174	22	2004 - 2026	8.0	-0.24	8.07	0	0
EB03	Decreasing trend	603667	23	2004 - 2026	7.9	-0.28	8	-0.01	0
EB04	No detectable trend	162510	6	2021 - 2026	7.8	-0.06	7.76	0	0.71

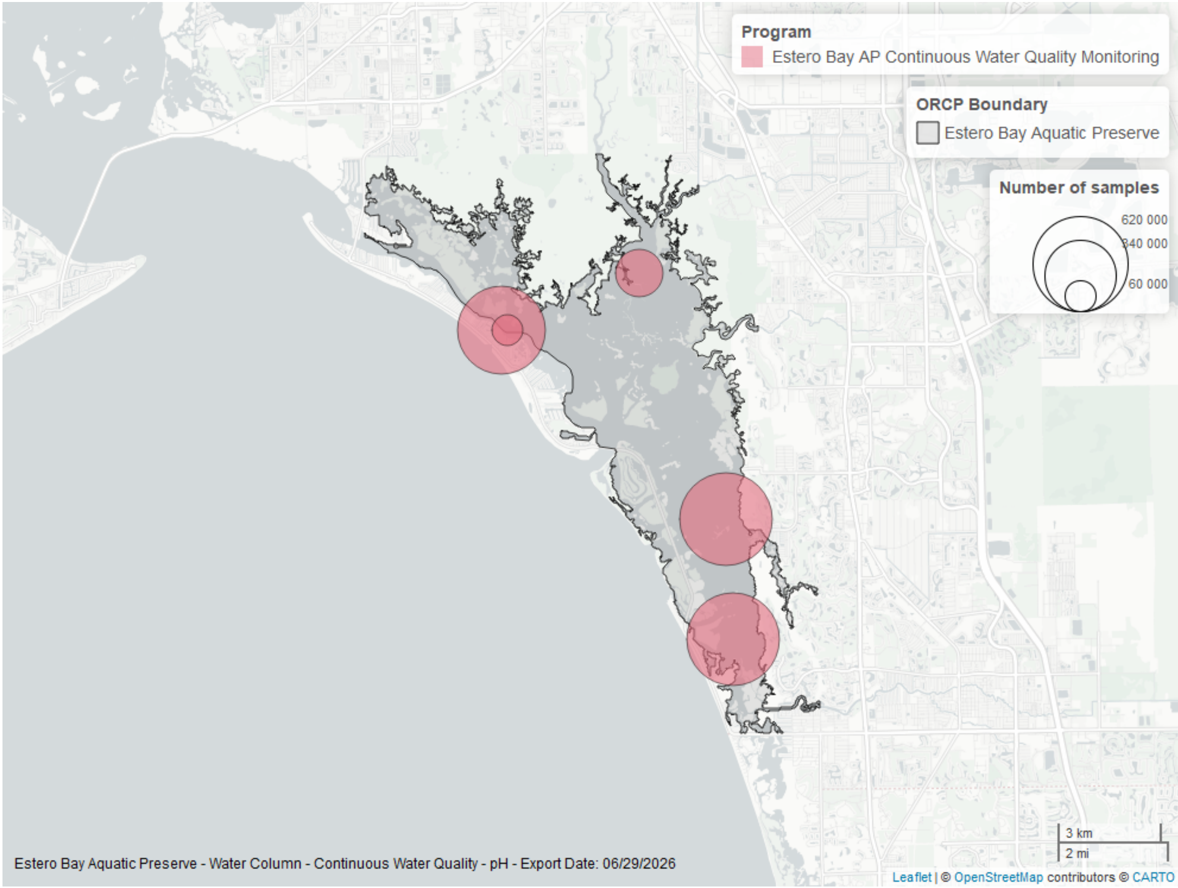

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

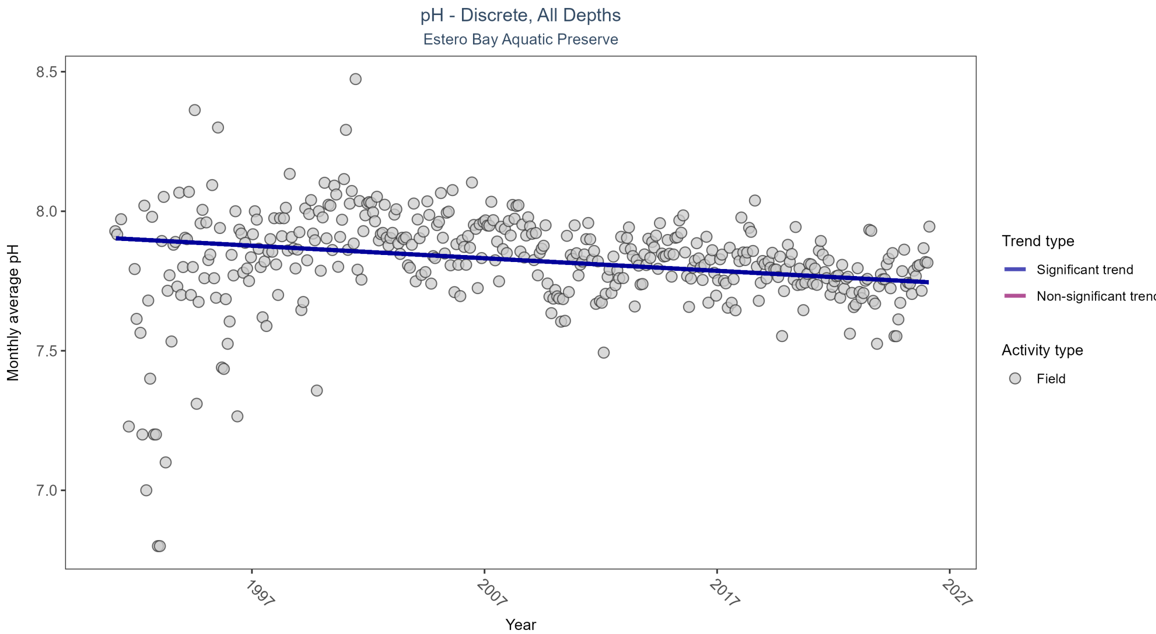

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	11160	36	1991 - 2026	7.9	-0.21314	7.90354	-0.0045	0

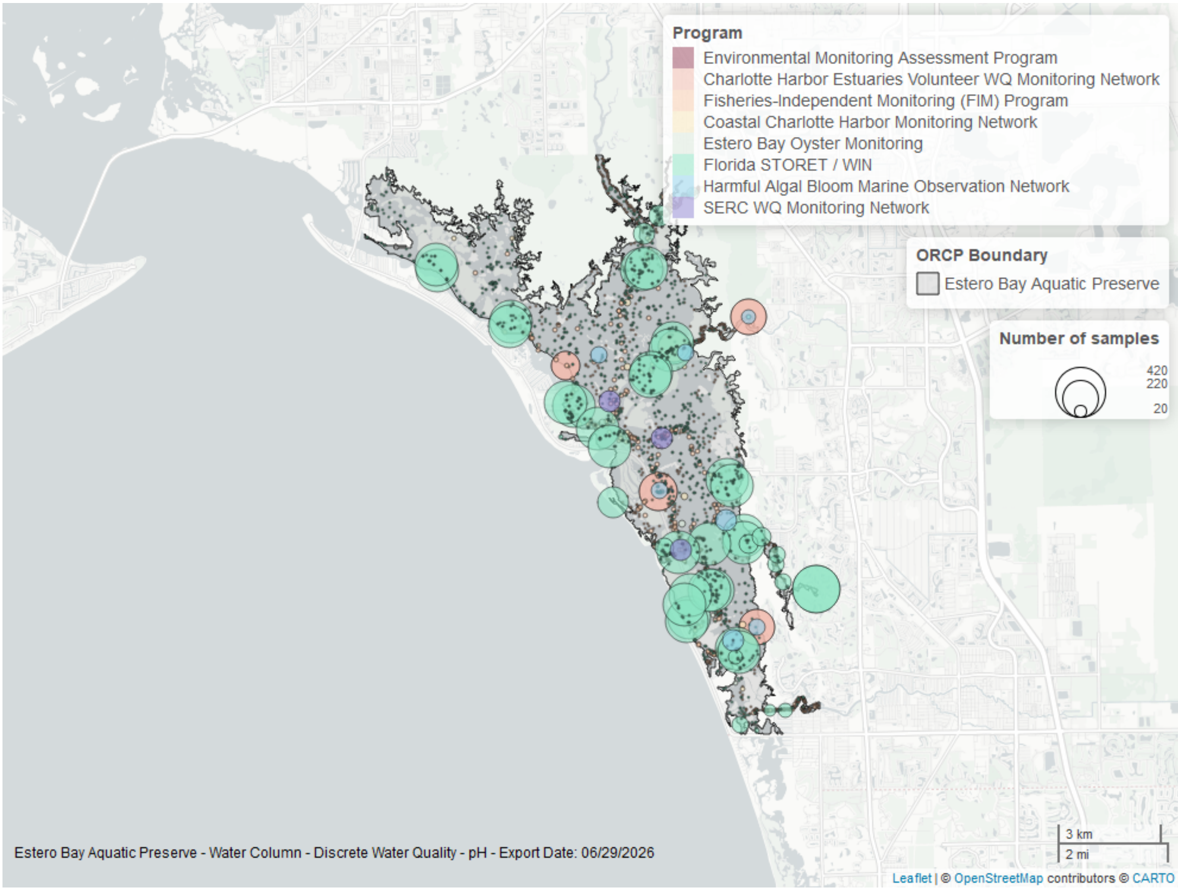

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

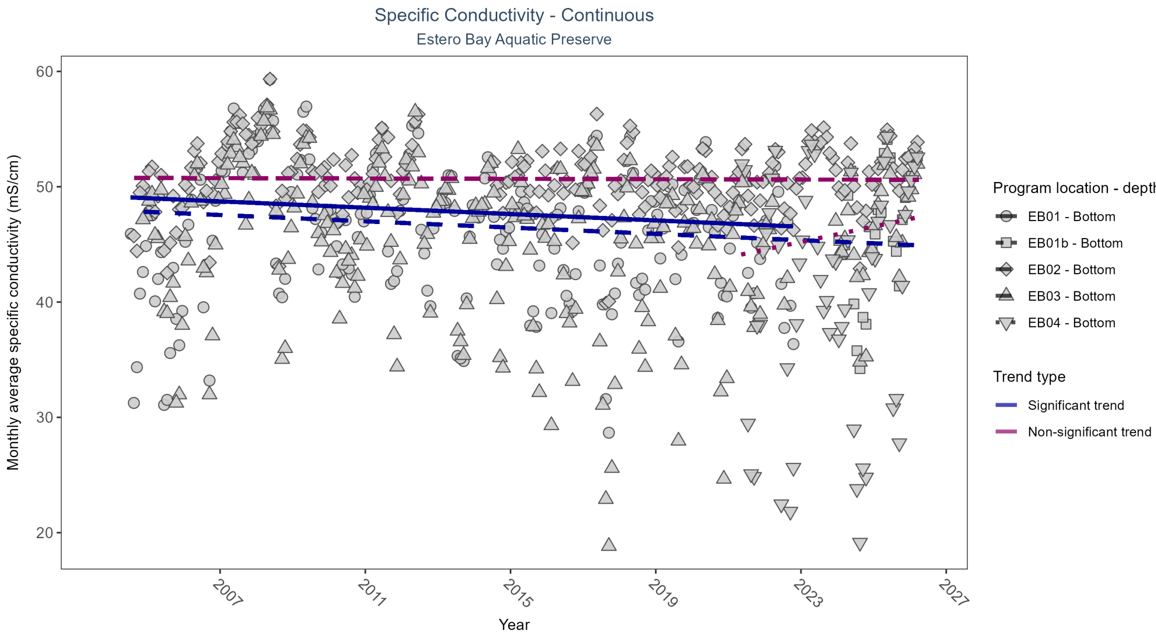

Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At two program locations, monthly average specific conductivity decreased by 0.14 mS/cm per year. No detectable change in monthly average specific conductivity was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Decreasing trend	564193	19	2004 - 2022	47.15	-0.13	49.14	-0.14	0.01
EB01b	Insufficient data	70428	3	2024 - 2026	46.98	-	-	-	-
EB02	No detectable trend	612803	22	2004 - 2026	51.31	-0.01	50.77	-0.01	0.81
EB03	Decreasing trend	605719	23	2004 - 2026	47.56	-0.14	47.96	-0.14	0
EB04	No detectable trend	160786	6	2021 - 2026	43.84	0.17	43.88	0.66	0.16

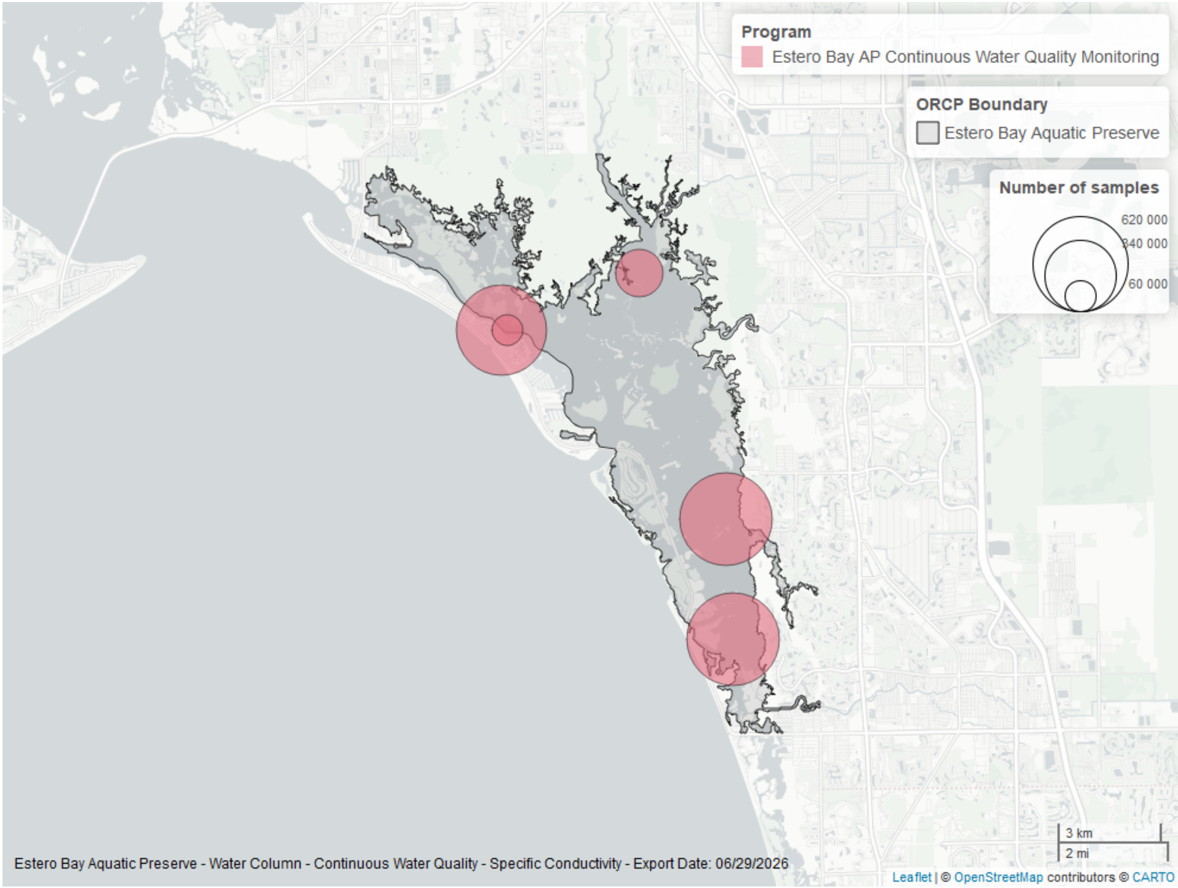

Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

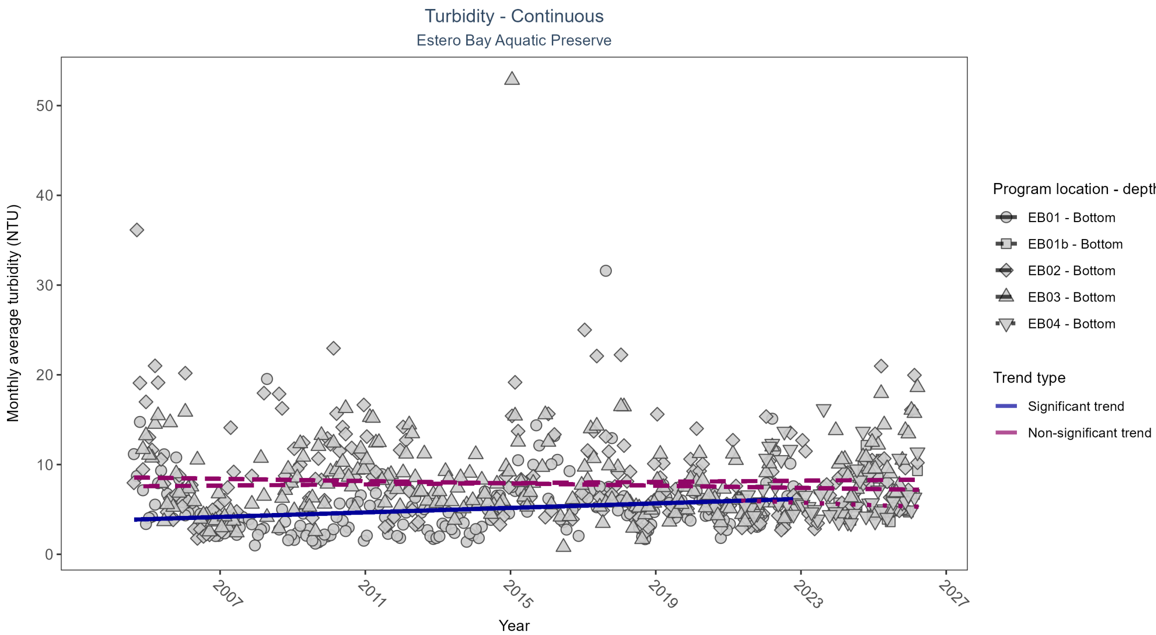

Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At one program location, monthly average turbidity increased by 0.13 NTU per year. No detectable change in monthly average turbidity was observed at three locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
EB01	Increasing trend	510965	19	2004 - 2022	4	0.18	3.79	0.13	0
EB01b	Insufficient data	72027	3	2024 - 2026	5	-	-	-	-
EB02	No detectable trend	511169	22	2004 - 2026	5	-0.09	8.6	-0.06	0.1
EB03	No detectable trend	491900	23	2004 - 2026	6	0.04	7.55	0.03	0.35
EB04	No detectable trend	155096	6	2021 - 2026	5	-0.14	6.04	-0.14	0.28

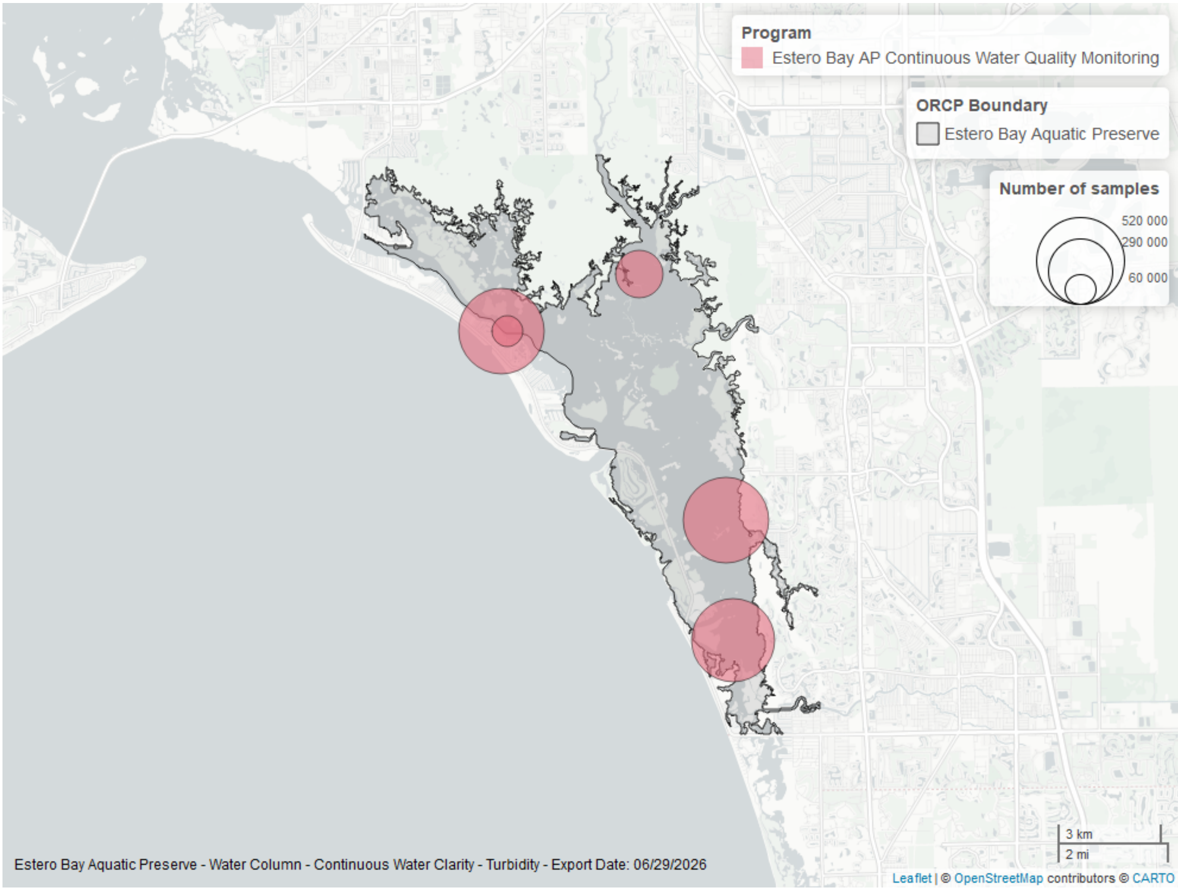

Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

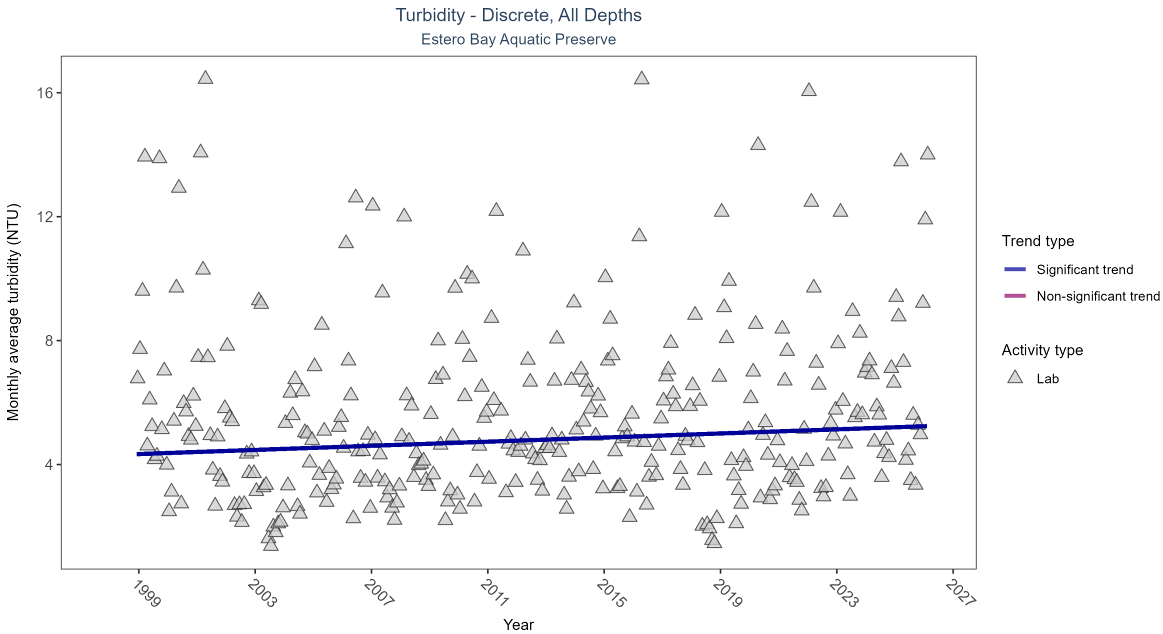

Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	3386	29	1998 - 2026	4.1	0.09139	4.30526	0.03321	0.0167

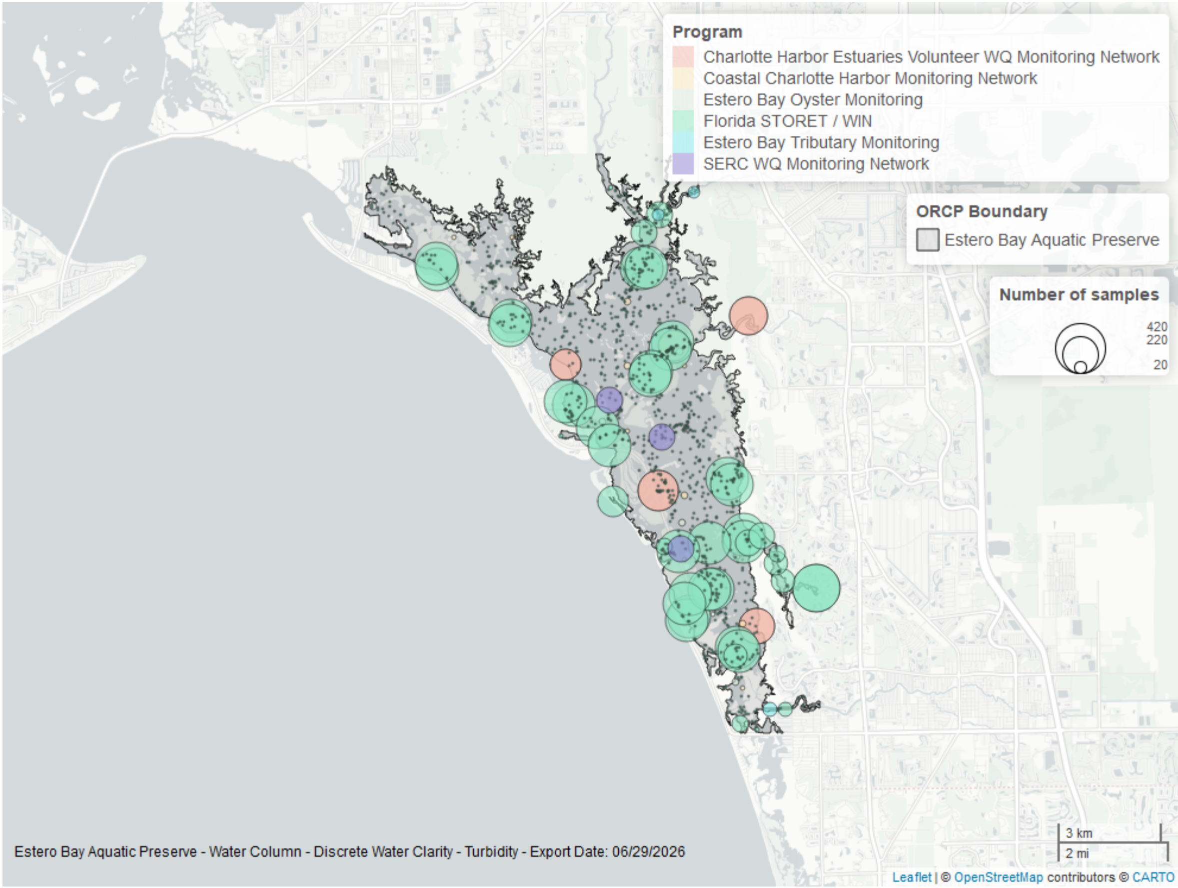

Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

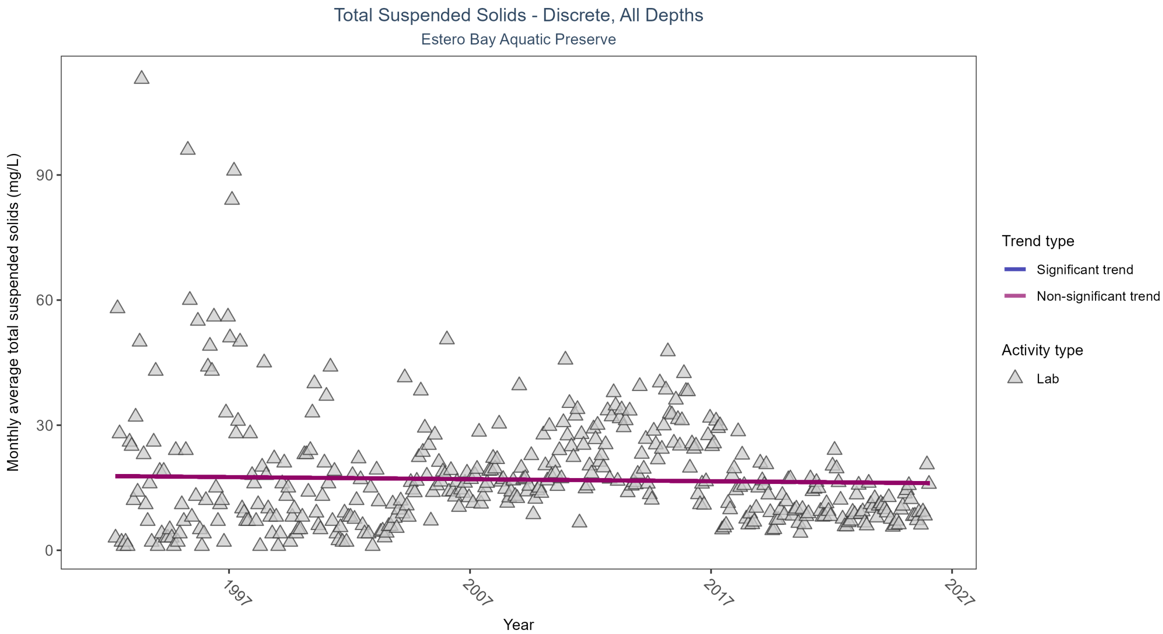

Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: Total suspended solids showed no detectable trend between 1992 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	5780	35	1992 - 2026	13.45	-0.03295	17.79848	-0.04966	0.3421

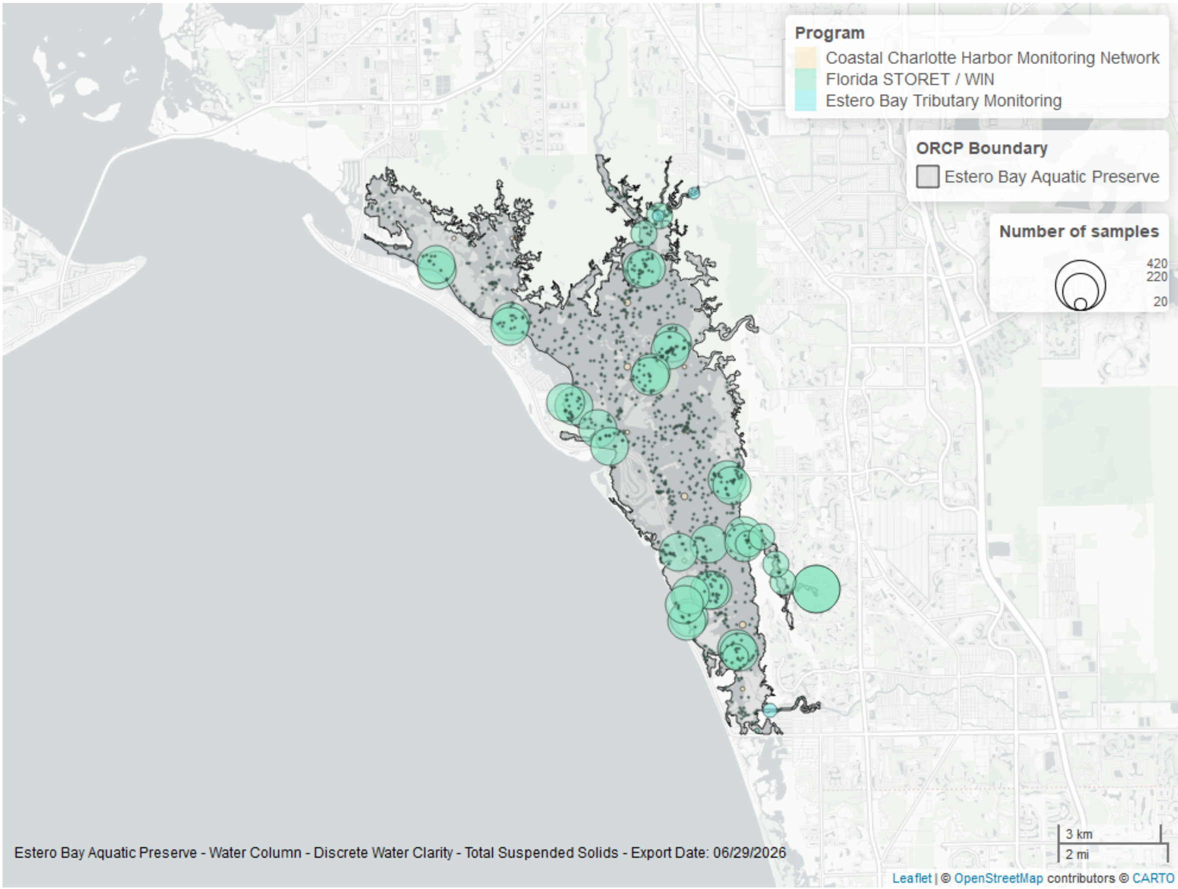

Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

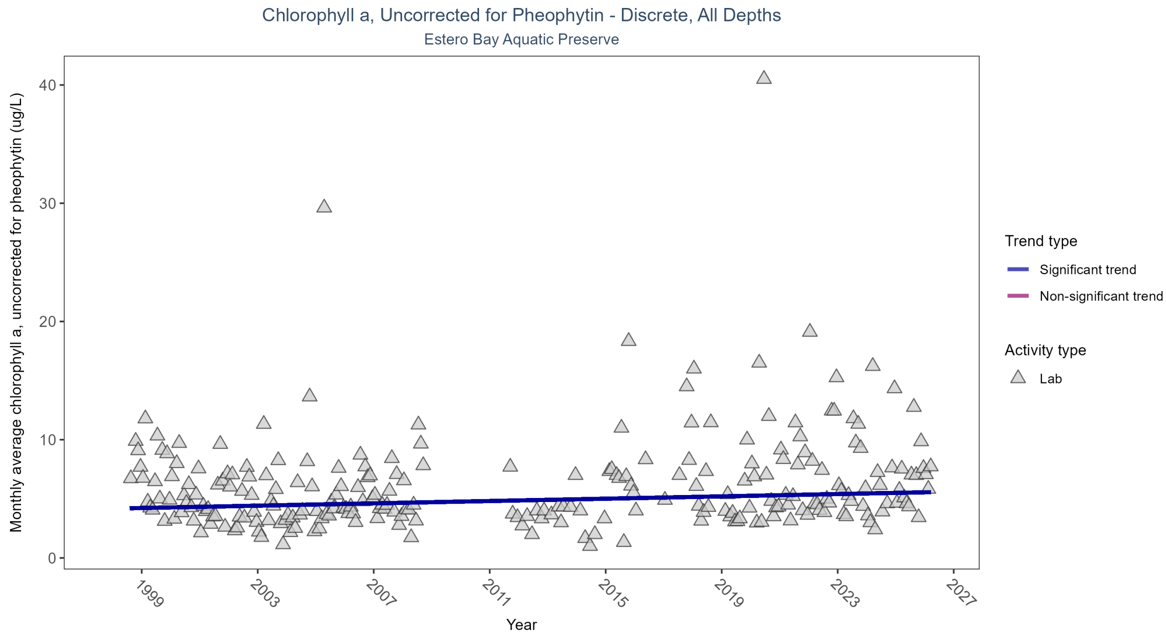

Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.05 ugg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1203	27	1998 - 2026	4.4	0.13956	4.17338	0.04919	0.0019

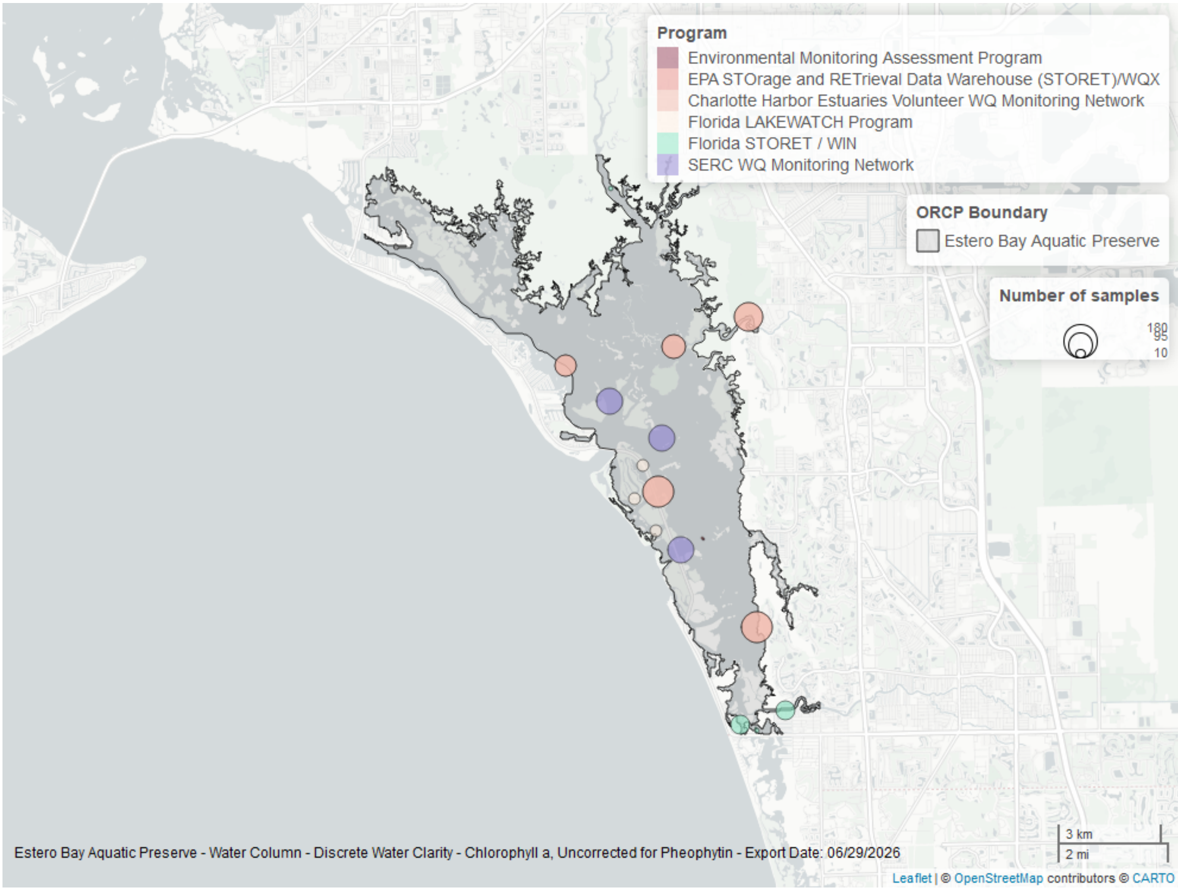

Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

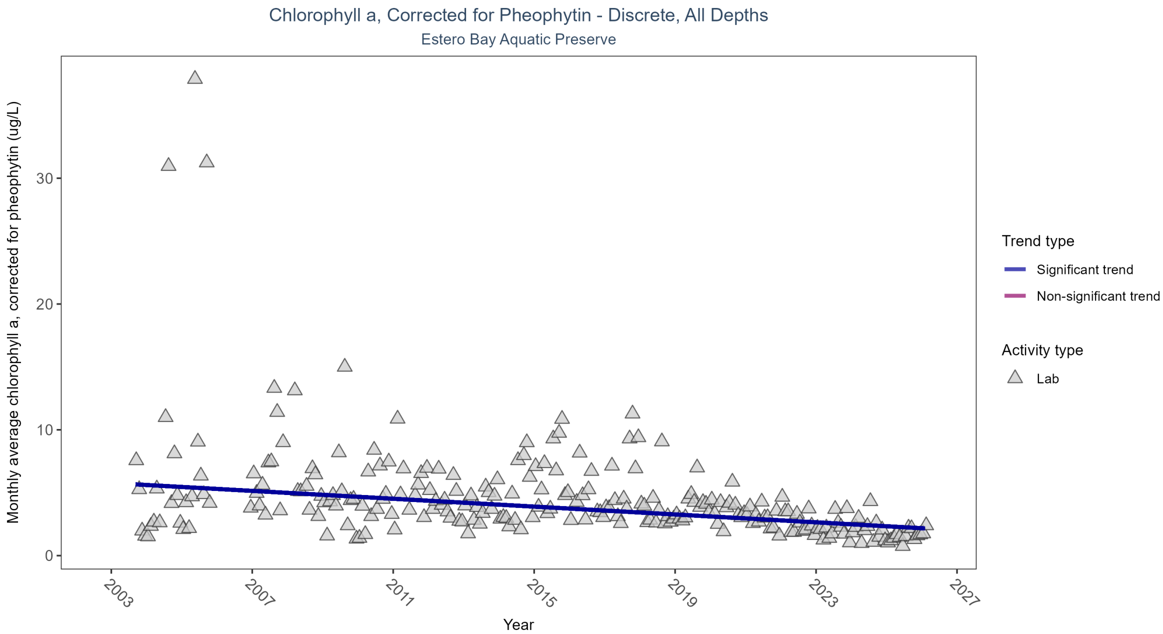

Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.16 ugg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2719	24	2003 - 2026	2.48	-0.35936	5.77036	-0.15616	0

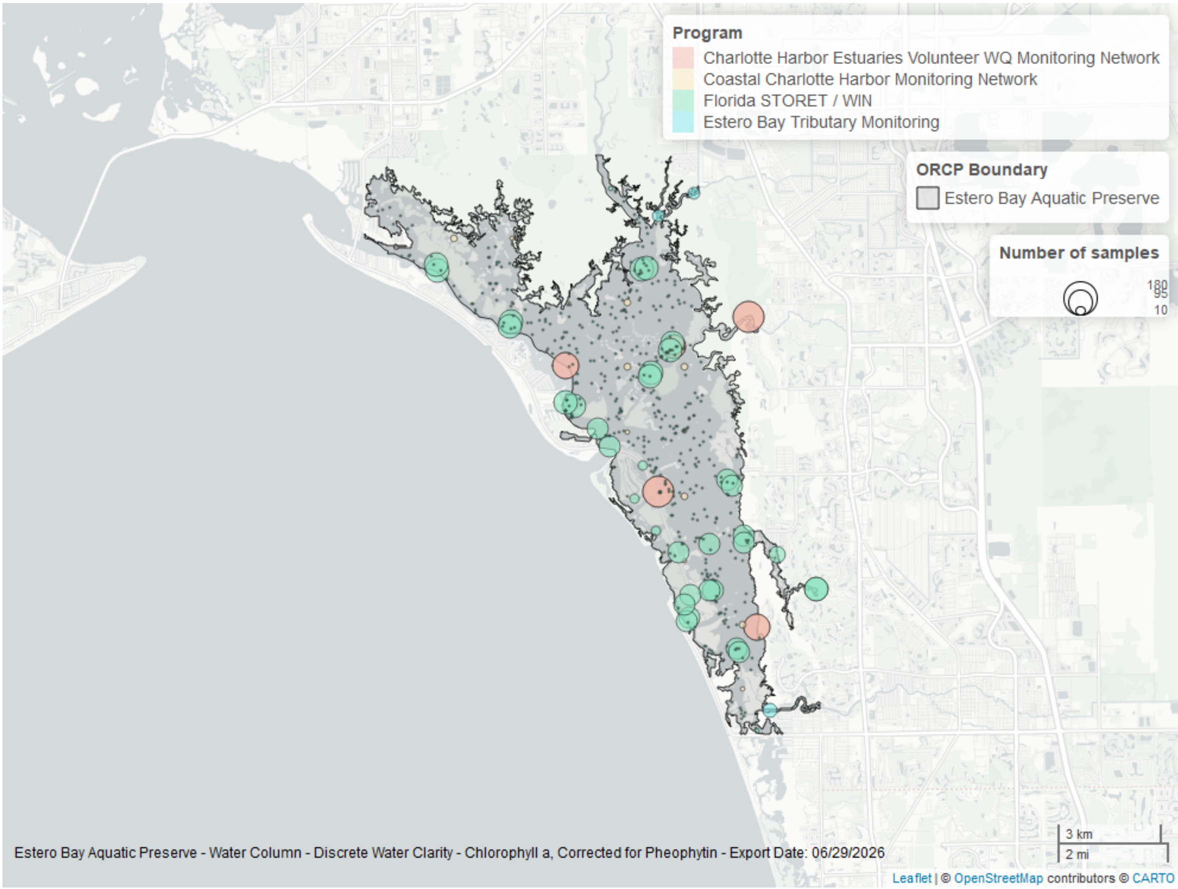

Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

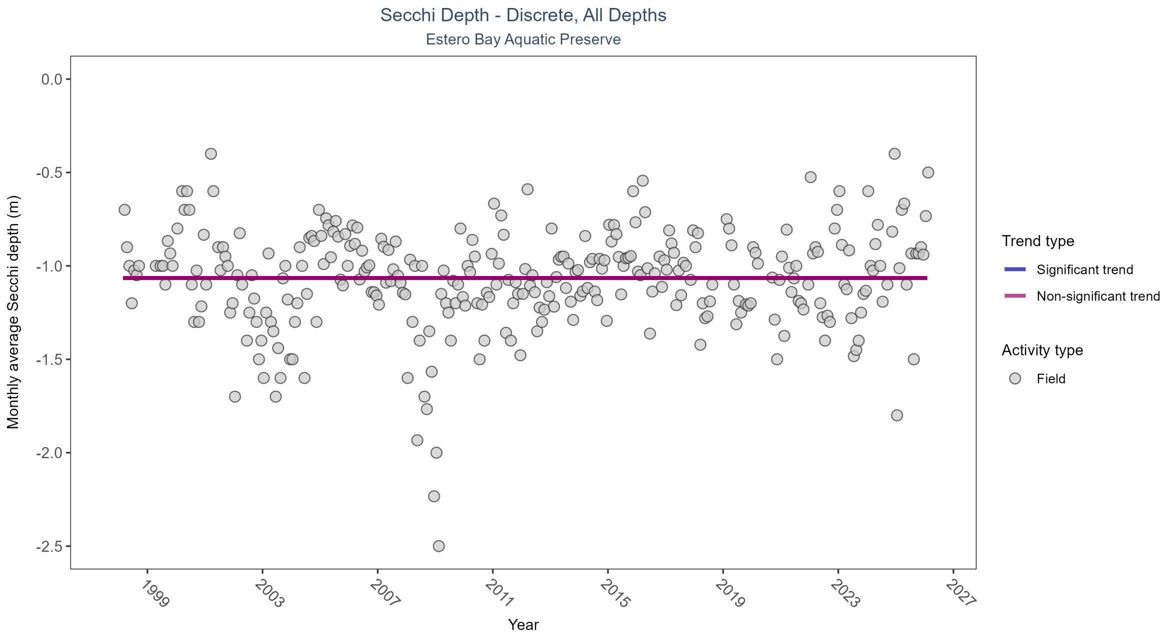

Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Secchi depth showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3373	29	1998 - 2026	-0.9	0.00256	-1.0652	0	0.9896

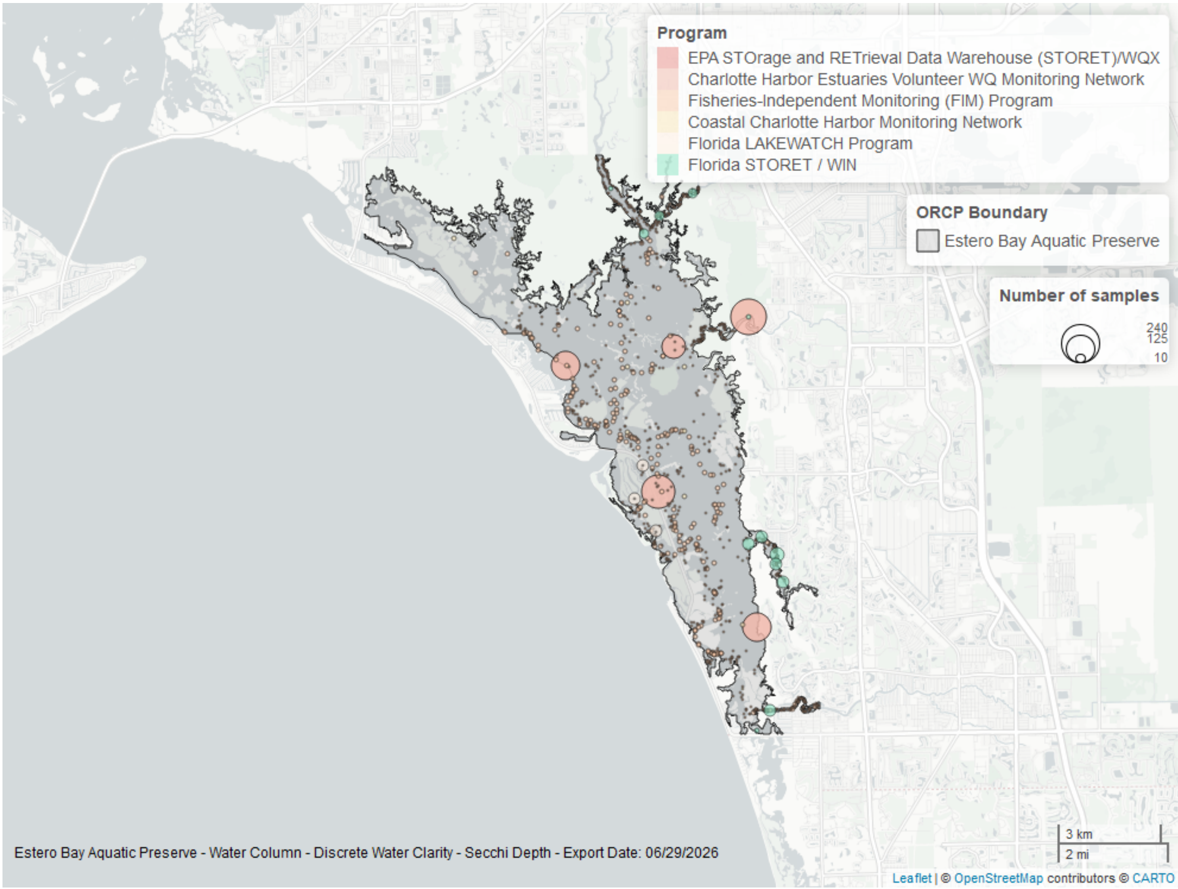

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

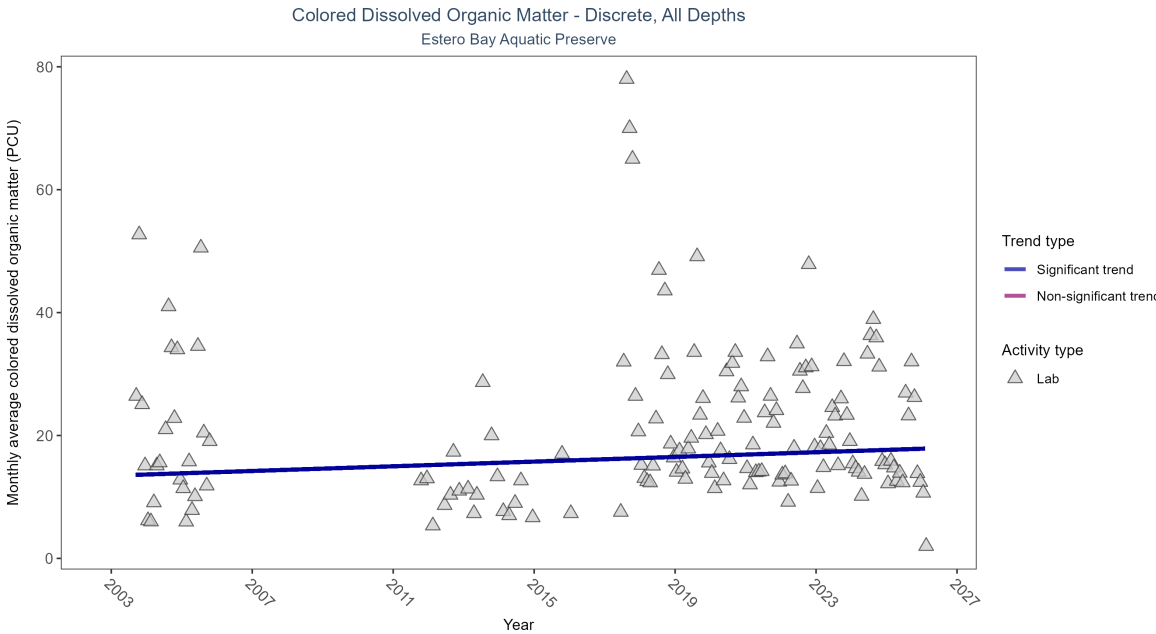

Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: Monthly average colored dissolved organic matter increased by 0.19 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2334	19	2003 - 2026	13.9	0.12156	13.46259	0.19057	0.0434

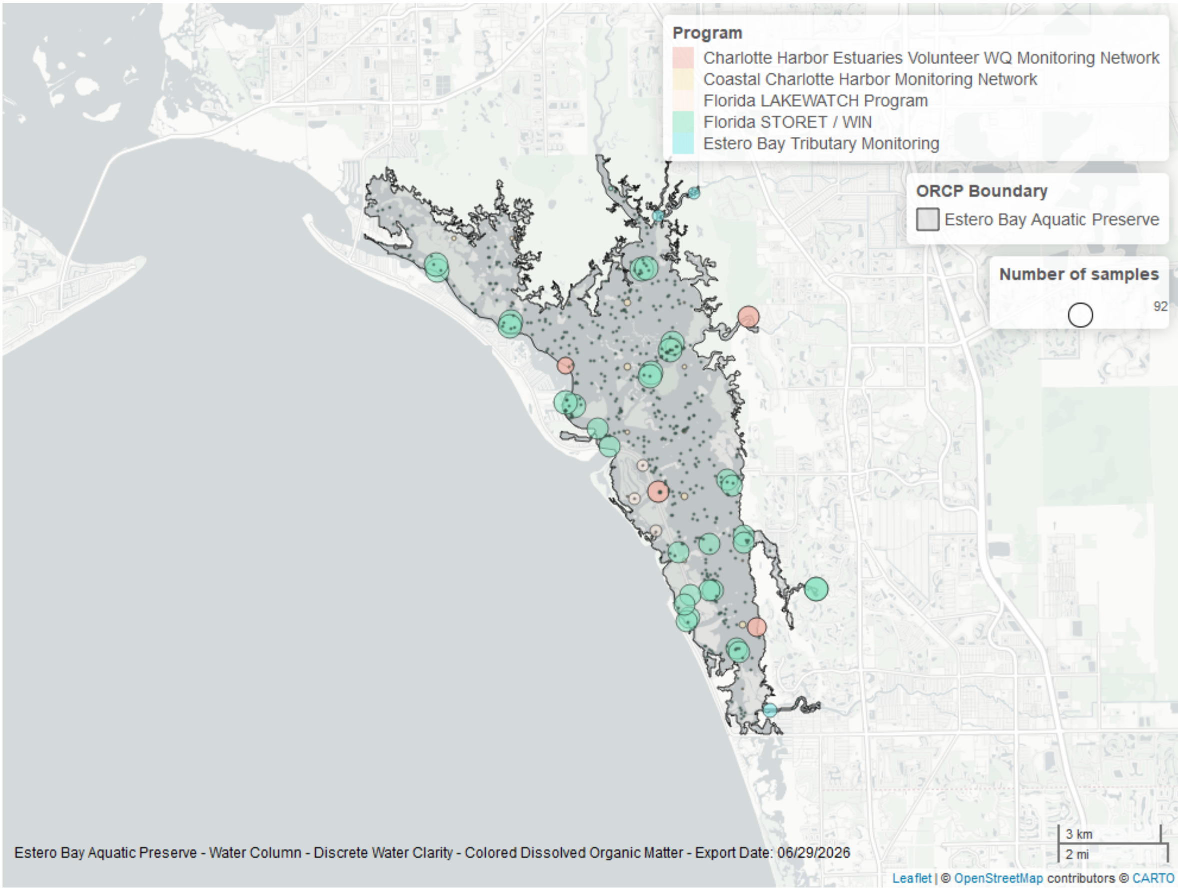

Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Estero Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.